



Creditor rights, collateral reuse, and credit supply☆☆☆

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ABSTRACT

Securities dealers receive mortgages as collateral for credit lines provided to mortgage companies and reuse the same collateral to borrow money. Exploiting the 2005 BAPCPA rule change, which granted mortgage collateral preferred bankruptcy treatment, I find that strengthening creditor rights increases dealers' collateral reuse. Increasing collateral reuse creates a money multiplier that increases credit supply. Using a novel dataset linking dealers to the mortgage companies they fund reveals that post-BAPCPA, dealers supply additional credit to mortgage companies by increasing credit lines and relaxing restrictions on collateral securing them. In response, mortgage companies increase origination volume and shift into riskier products.

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1. Introduction

According to theories of money creation, credit expands and contracts in the economy via a money multiplier in

the traditional banking system. My paper establishes a chain of lending in the sale and repurchase (“repo”) market that could potentially generate an analogous multiplier in the shadow banking system. This chain of lending arises when large securities dealers lend to mortgage companies in exchange for repo collateral and are allowed to reuse, or repledge, the collateral they receive. If dealers reuse the collateral to borrow from cash lenders and then reinvest the cash back into mortgage companies, it could lead to a money multiplier that originates in the repo market. While the money multiplier in the traditional banking system generates less money each round of lending, this repo multiplier could generate more money each round due to differences in the terms at which dealers and mortgage companies can borrow against the same collateral. Despite its potentially explosive effect on credit supply, the economic impact of this repo multiplier has not previously been explored. I exploit a change in repo creditor rights to shock dealer collateral reuse and trace the resulting supply of credit from dealers to mortgage companies and ultimately to mortgage originations.

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The mortgage companies that I study are independent mortgage companies (IMCs). Prior to the Global Financial Crisis (GFC), IMCs made up close to one third of the mortgage lending market. Post-crisis their market share has increased to approximately half of the market and their funding structure largely remains the same. Stanton et al. (2014) and Echeverry et al. (2016) establish that the IMCs rely heavily on credit line funding. However, there has been no direct evidence about who their funders were or how they operated. These questions have important implications for financial stability not only for the housing market but also for commercial real estate and collateralized loan obligations, which have adopted a similar funding structure.

To analyze who IMC funders were and how they operated prior to the GFC, I hand collect data on twelve of the largest public IMCs' credit lines between 2004Q3 and 2006Q3. I establish that these credit lines were collateralized by mortgage loans and implemented as *Master Repurchase Agreements* – contracts defining the purchase and resale of repo collateral – from the largest most interconnected repo dealers. I also collect data on the dealers' reported repledgeable collateral, or collateral eligible for reuse. Connecting a dealer's repledgeable collateral to the same dealer's funding to an IMC allows me to trace the transmission of an innovation in a dealer's repo collateral reuse to their increased credit supply to IMCs. I then use mortgage origination data to analyze the resulting impact on households.

I utilize Congress's passage of the Bankruptcy Abuse Prevention and Consumer Protection Act of 2005 (BAPCPA) to isolate a plausibly exogenous strengthening of creditor rights for repo mortgage collateral. The policy change exempted repo mortgage collateral from automatic stay. Importantly for my research design, BAPCPA only affected private-label, or risky, mortgage collateral since agency mortgage collateral had already been granted exemption from automatic stay by the Bankruptcy Amendments Act of 1984. Upon a bankruptcy event, exemption from automatic stay allows the final creditor to take immediate control of collateral without waiting in bankruptcy court. I hypothesize that this super senior bankruptcy status increased lenders' willingness to lend against the affected collateral and thus increased dealers' ability to repledge it. However, since private-label mortgage collateral is risky and repo transactions are typically backed by safe assets, it is not obvious that demand for this risky mortgage collateral would increase. Therefore, to empirically test whether BAPCPA increased collateral reuse and in turn credit supply, my analysis proceeds in three steps. First, I study whether the strengthening of creditor rights increases dealers' reuse of repo collateral; second, I study dealers' passthrough of their resulting credit supply increase to mortgage companies; and third, I study the mortgage companies' resulting increase in originations. Establishing consistent results at each step strengthens support for the underlying mechanism.

In the first step (Section 4), I test whether BAPCPA increases dealers' collateral reuse. To generate testable empirical predictions for collateral reuse, I first present a conceptual framework that illustrates the money multiplier

potential of reusing risky mortgage collateral. Following BAPCPA, my estimates show this multiplier was large – 4.5 times that of Treasuries. This large multiplier motivates my treatment intensity difference-in-differences research design to test whether strengthening repo creditor rights increases collateral reuse. More-treated dealers are those with a larger fraction of their pre-period balance sheet made up of collateral affected by BAPCPA. They would have a first mover advantage to generate a large money multiplier on a larger fraction of their balance sheet. Following BAPCPA, they could immediately redeploy the previously illiquid collateral by repledging it for cash to invest in more mortgages. If they engaged in multiple rounds of this cycle – lending to an IMC, receiving collateral from the IMC, repledging the collateral to a cash lender, reinvesting the cash by lending to an IMC – the multiplier would compound to persistently increase their credit supply. To test whether more-treated dealers enjoy this persistent increase in credit supply, I study the effect of treatment on dealers' reported repledgeable collateral following BAPCPA. I find that more-treated dealers experience an increase in repledgeable collateral relative to less-treated dealers following the change.

In the second step (Section 5), I test whether, following the policy change, treated dealers differentially increase their credit supply to IMCs relative to control dealers. Isolating the dealer credit supply channel requires simultaneously estimating both the dealer lending channel and the IMC borrowing channel, to control for mortgage demand confounders. A benefit of my data is that the same mortgage company receives funding from multiple dealers. This key feature allows me to study differential dealer lending within the same mortgage company. Following Khwaja and Mian (2008), I estimate a difference-in-differences analysis of the credit lines to a given mortgage company from treated versus control dealers. After BAPCPA, treated dealers increase their funding within the same mortgage company by 29.6% relative to control dealers. While this within-IMC, across-dealer, analysis is only able to establish a credit substitution effect between dealers, I also present evidence that suggests the policy change leads to a 19.9% overall credit supply increase to IMCs.

In addition, I find that post BAPCPA, dealers systematically relax covenants on their credit lines to mortgage companies. Rather than increasing funding for lines collateralized by traditional mortgages, dealers increase funding for “wet” (unsecured) credit lines and for lines collateralized by balloon, interest-only, and 120–180 day delinquent mortgages. This finding is consistent with BAPCPA increasing the expected recovery value of collateral, causing dealers to allow riskier types of mortgages to collateralize credit lines.

In the third step (Section 6), I conduct a treatment intensity difference-in-differences analysis to study whether IMCs pass the credit supply increase on to households. Due to data limitations that prohibit identifying individual IMCs, I utilize variation in the county-level market share of IMCs in 2004, the year before BAPCPA, to define counties more- versus less-exposed to the change. Prior to BAPCPA, I observe no statistically significant dif-

ference in mortgage volume and characteristics between counties with high- versus low-IMC market share. Post BAPCPA, a 10% increase in pre-period IMC market share leads to a 2.7% increase in mortgage originations during 2005–2006. The distribution of these originations shifts toward balloon and adjustable rate mortgages (ARMs), including interest-only ARMs; in line with dealers' increased funding for these products. These were “near prime,” or alternative, rather than subprime mortgage products. Consistently, I find that an increase in pre-period IMC market share drives an increase in the fraction of prime relative to subprime mortgages post BAPCPA. Although these alternative products boasted higher credit scores, which classified them as prime, Foote and Willen (2016) note that their structure may increase their risk of default. Consistently, within a five-month window around the shock, I estimate that the marginal default hazard rate increases from 13% pre-BAPCPA to 70% post-BAPCPA. A 10% increase in pre-period IMC market share leads to a 2.1% increase in home prices during 2005–2006 and to a 3.3% decrease in home prices during 2008, consistent with an amplification channel of repo funding. The results indicate that the credit supply increase in response to BAPCPA played an important role in the home price boom and its bust in 2008.

My paper contributes to the literature on collateral reuse in the financial markets. Infante (2019) models how dealers intermediate between repo lenders and borrowers using the same collateral. Fegatelli (2010); Singh (2011); Gottardi et al. (2019) establish that this kind of intermediation could lead to a money multiplier in the repo market. Jank and Moench (2020); Infante and Saravay (2020); Gorton et al. (2020) shed light on the multiplier of Treasury and agency repo post-GFC. My paper innovates relative to this literature by showing that the multiplier potential of private-label repo collateral is larger than that of safe assets. I also make a notable expansion to this literature by establishing the chain of lending that links the repo market to the housing market and by showing that an innovation to the repo multiplier is passed from the secondary market to the primary market.

My paper also contributes to the debate on whether repo should receive preferred bankruptcy treatment (Lubben, 2010; Roe, 2010; Skeel and Jackson, 2012). Supporters argue it improves stability of the financial markets (Edwards and Morrison, 2005). However, Duffie and Skeel (2012) propose four risks associated with granting repo preferred bankruptcy status: lowering repo lenders' incentive to monitor collateral; increasing their ability to become too big or interlinked to fail; increasing inefficient substitution toward short-term repo funding; and increasing risk of fire sales. My paper documents the first empirical evidence for each of these risks. Furthermore, one key concern preceding the Bankruptcy Amendments Act of 1984's expansion of repo safe harbors was that the repo collateral must be able to maintain its price in a crisis since granting it preferred bankruptcy status would facilitate fire sales. Consistently, Morrison et al. (2013) note that preferred bankruptcy treatment for repos was intended for collateral with price stability. However, Maclachlan (2014) notes that a discussion of whether

mortgage collateral could maintain its price in a crisis was largely neglected prior to BAPCPA's expansion of repo safe harbors. The findings in my paper underscore the importance of verifying collateral's ability to retain its price in a crisis before granting it exemption from automatic stay.

Additionally, my paper sheds light on three puzzles surrounding the GFC. First, how did the relatively small realized losses found in Ospina and Uhlig (2018) on mortgage-backed securities put 12 of the 13 most systemically important financial institutions at risk of failure in a period of two weeks? Rajan (2005) and Acemoglu et al. (2015) warn that critical interlinkages exposed the economy to low probability events, yet the exact role played by the financial system's architecture in creating systemic risk remains imperfectly understood. My paper highlights the role of repo rehypothecation, or the reuse of collateral, in increasing critical interlinkages that were backed by risky collateral and exposed to cross-default clauses leading up to the GFC.

Second, was the run in the repo market – failure to roll over short-term repo loans – central to the GFC as proposed in Gorton and Metrick (2010a, 2010b, 2012)? While Krishnamurthy et al. (2014) find evidence of a run, they find that only a small fraction of total repo, repo backed by mortgage collateral, was exposed to the run. My paper illuminates how BAPCPA increased dealer exposure to repo runs on their liability side and the kind of collateral runs described in Infante and Vardoulakis (2021) on their asset side; allowing a run in even a small segment of repo to have devastating effects. Consistently, Singh and Aitken (2010) show that repo collateral reuse increased dealer leverage by 50% more than standard estimates during 2007–2009.

Third, was the build up in home prices leading up to the GFC triggered by credit supply or expectations (Mian and Sufi, 2009)? I document evidence consistent with a credit supply expansion. I also shed light on why 2006–2007 vintage mortgages defaulted faster than 2000–2004 vintages, as found in Demyanyk and Van Hemert (2011), and why the majority of mortgage defaults were focused in the prime rather than subprime segment, as found in Albanesi et al. (2017).

There is an existing literature that uses BAPCPA's repo safe harbor provision as a natural experiment. Srinivasan (2017) shows that BAPCPA increased demand for private-label mortgage collateral in one segment of the repo markets. Bellicha (2016) and Ganduri (2016) study the impact of BAPCPA on the deterioration of mortgage originations. Chircop et al. (2018) find that information asymmetry increased for banks exposed to BAPCPA's repo provision. I innovate relative to this literature by studying BAPCPA's effect on dealers' reuse of mortgage collateral in the repo markets and the credit creation that it leads to in the real economy.

2. Data

My empirical analysis combines dealer and mortgage company borrowing and lending data.

2.1. Independent mortgage companies

The credit lines IMCs used to fund mortgage originations were called warehouse facilities. To select the sample of IMCs whose warehouse facilities I collect, I narrow the IMCs to public companies. Most IMCs in the sample become public in mid-2004, thus I observe their data beginning in third quarter of 2004. I choose the IMCs that report the dealers from whom they borrow in their quarterly and annual financial statements filed with the SEC from 2004Q3 to 2007Q3. This gives me a sample of 12 IMCs. Each IMC reports the dealers funding each of its credit facilities and the maximum amount of the facility each quarter. After 2006Q3 seven of the IMCs filed for bankruptcy or were acquired, so that I am no longer able to pull their quarterly financial statements.¹ Thus, I limit my sample to 2006Q3.

The data capture credit lines reported as warehouse lines of credit, warehouse repurchase facilities, and repurchase agreements. A subset of these mortgage companies also report their utilization on credit facilities and the posted mortgage collateral, allowing me to calculate their overcollateralization. I provide an example of the data that I collect in Appendix A in the Online Appendix and present evidence that the warehouse repurchase facilities were structured as Master Repurchase Agreements in the bilateral repo market. For the remainder of the text, I will refer to these as Master Repurchase Agreements, warehouse repurchase facilities, or credit facilities.

The IMCs were 12 of the largest public IMCs and generated 59% of all mortgages originated by IMCs in 2006, as discussed in subsection B.3 in the Online Appendix. I calculate that the IMCs' main source of funding is from warehouse repurchase facilities funded by dealers making up 61% of IMC assets on average.

2.2. Dealers

Dealer Repledgeable Collateral. There is limited data on the repo markets prior to 2008 (Baklanova et al. (2015)). Therefore, I use IMCs' warehouse lines to link IMCs to the dealers who were lending to them. During 2004Q3–2007Q3, I find that 29 dealers were funding the IMCs and 16 of the 29 were primary dealers. The primary dealers are a subset of broker dealers who deal directly with the government to make the market for newly issued US Treasuries. They are the most interconnected broker dealers. Once I limit the dataset to 2006Q3, the number of dealers falls from 29 to 27, and the number of primary dealers falls from 16 to 15, as discussed in subsection A.5 of the Online Appendix.

To study reuse of repo collateral by the dealers funding the IMCs, I collect dealers' reported repurchaseable collateral for 19 of the 29 dealers – the 16 primary dealers and 3 additional dealers. Dealers report in a footnote of their annual reports the amount of collateral that they receive which is permitted to be re-sold or repurchased. I col-

lect these data from year end 2000 through year end 2008, with varying start dates. Year end 2004 is the earliest date in which all dealers report. These data allow me to study the amount of collateral that dealers receive which they are permitted to repledge pre- versus post-BAPCPA. I report the dealers and the data collected for them in subsection A.5 of the Online Appendix.

Dealer Repo Financing Data. In addition to the dealers' repurchaseable collateral data, I use the Federal Reserve Bank of New York's weekly survey of primary dealers (FR 2004) to measure primary dealers' aggregate trading activity by collateral class. Due to their role making the market for US Treasuries, the primary dealers are required to report trading data to the Federal Reserve. There is no external data dictionary identifying financing positions and collateral classes for these data outside the Federal Reserve. To overcome this, I hand match the FR 2004 variables to the survey instructions given to the survey respondents in order to create a weekly time series of total dealer holdings and secured financing by collateral class. Matching the FR 2004 variables to the survey instructions allows me to establish that prior to 2013 the data included dealers' trading in private-label mortgage collateral in the line item *corporate securities*, as discussed in subsection A.1. In the Online Appendix, I decompose corporate securities into the collateral classes that comprise it.

These data allow me to study whether primary dealers' trading in the repo markets using private-label collateral increased post BAPCPA. The key variables that I utilize are *securities out* and *securities in*. *Securities out* reports dealers' secured borrowing (cash received) and *securities in* reports dealers' secured lending (cash lent).² The FR 2004 data report the primary dealers' aggregate activity in both the tri-party and bilateral repo markets. Copeland et al. (2014) states that primary dealers made up 79% of all dealer activity in the tri-party repo market in July and August 2008 and assumes that this percentage holds across both the bilateral and tri-party repo markets. Therefore, the FR 2004 trading activity is likely to be representative of trading activity in the repo markets as a whole during 2005–2007.

Securitization and Price of Mortgage Backed Securities (MBS). This paper leverages data on dealer securitization of private-label mortgage backed securities (MBS) from CoreLogic ABS database and Inside Mortgage Finance's Mortgage Market Statistical Annual. These data allow me to identify dealers who were more heavily exposed to private-label MBS (PLS) securitization and therefore warehouse private-label mortgage collateral in 2004, the year prior to BAPCPA. They capture the value, year, and collateral class of deals underwritten by each dealer. Using these data, I compute the value of subprime residential MBS deals securitized in 2004 by each dealer in my sample. In 2004, subprime deals comprised the lion's share of the private-label market. The private-label market generally is comprised of "subprime" and "near prime" mortgages (Adelino et al., 2019). Alternative A-paper (Alt-A) or

¹ For example New Century formally filed for bankruptcy in April 2007. ECC Capital Corp was purchased by Bear Stearns and this sale was closed in February 2007.

² *Securities out* and *securities in* include repos/securities lending and reverse repos/securities borrowing, respectively. See: FR 2004 and Infante (2019) p. 46.

“near prime” deals increased in response to BAPCPA, as discussed in Section 6.

In order to study the effect of BAPCPA on the price of PLS in the secondary market, I study the daily average yields on the LD100AS Bloomberg Barclays agency MBS index and the BNA10AS Bloomberg Barclays private-label MBS index from October 2003 through December 2006.

2.3. Mortgage market data

To establish the effect of BAPCPA on IMCs' lending to households, I utilize the Home Mortgage Disclosure Act (HMDA) data, as well as the CoreLogic data.

HMDA Data. In order to supervise and enforce fair lending practices nationwide, the U.S. Congress mandates that all loan applications related to home purchase, refinancing, and home improvement be reported to the federal government. The main variables that I use from these data are whether a mortgage was originated, who the originator was, whether the originator was an IMC, the year, and county in which it was originated. I use these data to construct the IMC county level market share in 2004, the year prior to BAPCPA. To identify the IMCs, I use the crosswalk maintained by Robert Avery to match subsidiaries belonging to the same parent company and I aggregate mortgages originated by each parent company. I define a mortgage company as an IMC if it underwrites and funds a loan in its own name, following the HMDA definition of IMCs. I also utilize the county, month data provided by Neil Bhutta in order to study granular time variation around BAPCPA, as the public HMDA data only publishes data at the annual level.³ At the county month level, the HMDA data no longer tracks individual IMCs.

CoreLogic Data. I use the CoreLogic Loan Level Market Analytics (LLMA) data to study mortgage characteristics and originations pre- and post-BAPCPA. Due to data restrictions, I am not able to observe the originator of a mortgage or to identify whether or not it is an IMC. Therefore I am not able to compare the individual mortgages originated directly by my treated versus control IMCs, or other institutions, pre- and post-BAPCPA. To overcome this, I aggregate all variables to the county level and merge with the pre-period IMC county level market share to analyze the effect of exposure to IMC lending on changes in loan characteristics.

The LLMA contain detailed information on mortgage characteristics at origination as well as monthly performance data for a large sample of anonymized borrowers. CoreLogic collects these data from 25 of the largest mortgage servicers in the U.S. The LLMA data track approximately 5.7 million mortgages each year and in a typical year include 45% of mortgages originated in the U.S. during 2003–2008. The main variables that I utilize are the mortgage's initial interest rate, occupancy status, mortgage product (balloon, negative amortizing, adjustable rate mortgage (ARM)), and prime versus subprime status. I use the monthly performance data over the life of a mortgage

to study the effect of BAPCPA on likelihood of default. I use the variable “mba_delinquency_status,” reported in accordance with the Mortgage Bankers' Association (MBA) standards, which records the status of a borrower's payments on the loan and provides indicators for foreclosure, bankruptcy, and real estate owned (REO) properties. REO properties are properties seized by lenders from borrowers who are unable to pay their mortgages.

Home Price Data. In order to study the effect of BAPCPA on home prices at the county level, I use the county level Zillow Home Value Index (ZHVI). ZHVI is a time series tracking the monthly median home value in a particular county across the sample period. I utilize these data to causally identify the effect of a credit supply expansion in the repo markets on home prices.

3. Institutional background

In order to understand the research design and the credit supply increase that BAPCPA generated, this section describes (1) the structure of IMC credit lines from dealers; (2) how dealers operate in the repo market; (3) how BAPCPA affected the interactions between dealers and IMCs.

Independent Mortgage Company (IMC) Warehouse Repurchase Facilities. Independent mortgage companies are not deposit taking institutions. They depend heavily on the sale of their mortgages in order to fund themselves. This sale typically takes between 30–60 days. In the meantime, mortgage companies originate mortgages, package them into a warehouse facility and use this warehouse facility to borrow against. The repayment of these warehouse credit lines varied by contract but they were often repayable either when the financed loans were sold or on the maturity date of the contract.⁴

Repo Market Functioning. The repo markets are large short-term funding markets where securities are sold and repurchased, creating short-term loans collateralized by financial assets.⁵ Copeland et al. (2014) estimate that during July–August 2008, the sum of all repos outstanding on a typical day was approximately \$6.1 trillion. The sum of all reverse repos outstanding was about \$4 trillion.⁶ The main users of repos are large dealer banks and other financial institutions such as money market funds (MMFs), hedge funds, and IMCs.

The repo market consists of two segmented markets: the **bilateral** and the **tri-party** market. Lower overcollateralization rates or “haircuts” are charged in the tri-party market on the same collateral, due to lower counterparty risk. This is because traditionally more credit worthy market participants – such as large dealers and cash investors – trade in the tri-party market, the collateral is held by

⁴ See: HomeBanc 2005 10-Q3 p 101 of 173.

⁵ For a succinct description of repos see Bevil (1989) 878 F.2d 742, 743 (3d Cir. 1989).

⁶ About 40% of repo activity was in tri-party repos and the remaining 60% was in bilateral repos. About 92% of reverse repos took place in the bilateral market. Due to double counting, summing the total repo and reverse repo values may overstate the total size of the market. Copeland et al. (2014) p. 2348.

³ For all specifications, I limit the data to the top 500 counties captured in the dataset published by Neil Bhutta: <https://www.sites.google.com/site/neilbhutta/data>.

a clearing house (the third party), and the contracts are short-term in nature. The clearing house provides several important roles, such as settling and netting transactions, and the bankruptcy treatment of collateral affects the clearing house's ability to perform these functions. In the event of bankruptcy, the automatic stay would interfere with the timely settlement of collateral held in the clearing house's custodial accounts. The bilateral market is where opaque, less credit-worthy agents seek short-term funding. The cash borrowers in this market are riskier and face larger haircuts to protect the dealers lending to them.

The tri-party market is the market that connects dealers with nonbank cash investors such as MMFs and securities lenders. Krishnamurthy et al. (2014) state that the tri-party market is the way in which cash funding enters the shadow banking system through repo. The bilateral repo market is a market through which funds are reallocated between dealers and across dealers and hedge funds, or mortgage companies. Dealers play an important role as repo intermediaries between cash lenders and cash borrowers across these markets. Dealers are likely to be cash borrowers (send securities out), in the tri-party market (Infante (2019)), and to be cash lenders (receive securities in) in the bilateral market (Copeland et al. (2014)). Tuckman (2010) states that dealers prefer to use borrowed capital to finance their borrowing rather than to use their own scarce capital. Thus dealers repledge or rehypothecate collateral received in the bilateral market, in the tri-party market (Copeland et al. (2014)). This reuse of collateral allows dealers to take advantage of the differential between haircuts in the bilateral and tri-party markets to generate a cash "windfall" (Infante (2019)).

Bankruptcy Abuse Prevention and Consumer Protection Act of 2005 (BAPCPA). Repurchase agreements using collateral defined in the bankruptcy code receive exemption from automatic stay. This grants the holder of the underlying collateral super-senior bankruptcy status since the collateral is exempt from the hold on a firm's assets when the firm enters bankruptcy proceedings. However additional types of collateral, not explicitly defined in the bankruptcy code, are also traded in the repo markets. All repurchase agreements are written with the standard contract in the hopes that the court will interpret them as receiving preferred bankruptcy status. However, the preferred bankruptcy status relies on the court's interpretation (Lumpkin (1993)). The market contraction in response to two important bankruptcy court cases where the court failed to grant repo collateral preferred bankruptcy status, Lombard Wall (1982) and Criimi Mae (2000)– both heavily funded with repos – underscores this point. The market response to these court cases suggests that collateral must be legally exempt from automatic stay in order for cash lenders in the tri-party market to lend against it. I discuss these court cases further in the Appendix A in the Online Appendix.

BAPCPA was introduced in Congress in February 2005 and signed into law in April 2005.⁷ The law expanded

the definition of repurchase agreements in the Bankruptcy Code to include: (1) mortgage loans; (2) mortgage-related securities; (3) interests in mortgage-related securities or mortgage loans. This granted private-label mortgage securities and whole loans exemption from automatic stay, giving the final creditor super senior bankruptcy status. This expansion only affected private-label mortgage collateral, since agency mortgage collateral had been exempted in 1984. I hypothesize that this enabled the clearing house in the tri-party market to hold the collateral without worry that a counterparty's failure would trigger automatic stay.

These institutional details suggest that BAPCPA enabled dealers to reuse, or repledge, warehoused private-label mortgage loans or newly minted private-label MBS in the tri-party market in larger quantities than they had been able to prior to the law change. In the sections that follow, I present evidence consistent with this hypothesis from analyses on dealers' collateralized repo borrowing and MBS yields. In Appendix A in the Online Appendix, I provide evidence that participants in the tri-party market increased their lending against mortgage collateral, however, the mechanism will go through even if BAPCPA only increased dealers' reuse of private-label mortgage collateral in the bilateral repo market.

Conceptual Framework. This section sets up the conceptual framework that depicts how the money multiplier works and generates predictions, which I empirically test in the following sections. I propose that prior to BAPCPA, reuse of the mortgage collateral was limited. The collateral was simply held – with limited reinvestment capability – to protect dealers from the default risk of the mortgage companies. Following BAPCPA, I propose that dealers' reuse of mortgage collateral expanded, allowing them to generate a money multiplier by taking advantage of the differential between the haircut that they charged in the bilateral market relative to the haircut that they received in the tri-party market.

In Fig. 1, I depict a stylized example of the proposed change in dealers' ability to reuse the warehoused mortgage loans posted by an IMC prior to BAPCPA, in panel (a), versus after BAPCPA, in panel (b).⁸ To simplify the exposition of the money multiplier, in (a) I depict the tri-party market refusing to lend against warehoused mortgage loans, corresponding to a 100% haircut. In (b) I propose that BAPCPA increased dealers' ability to reuse the collateral. This argument would go through if dealers were able to repledge whole mortgage loans from the mortgage warehouse, or if they were able to securitize the

2005, and signed into law by the president of the U.S. on April 20, 2005 (<https://www.congress.gov/bills/109th-congress/senate-bill/256/text/enr>). It applied to consumer bankruptcy cases after October 17, 2005. See: Bankruptcy Abuse Prevention and Consumer Protection Act of 2005, Pub. L. No. 109-8, §907, 119 Stat. 23, 171–172 (codified as amended at 11 U.S.C. §101(47) (2012))

⁸ Although Fig. 1 (b) and the argument below focus on reuse of warehoused mortgage collateral in the tri-party market, the argument would go through if dealers increased reuse of collateral in the bilateral market at lower haircuts than they received prior to the policy change.

⁷ The law was introduced in Congress on February 1, 2005 by Republican Senator Chuck Grassley, passed by Congress on April 14,

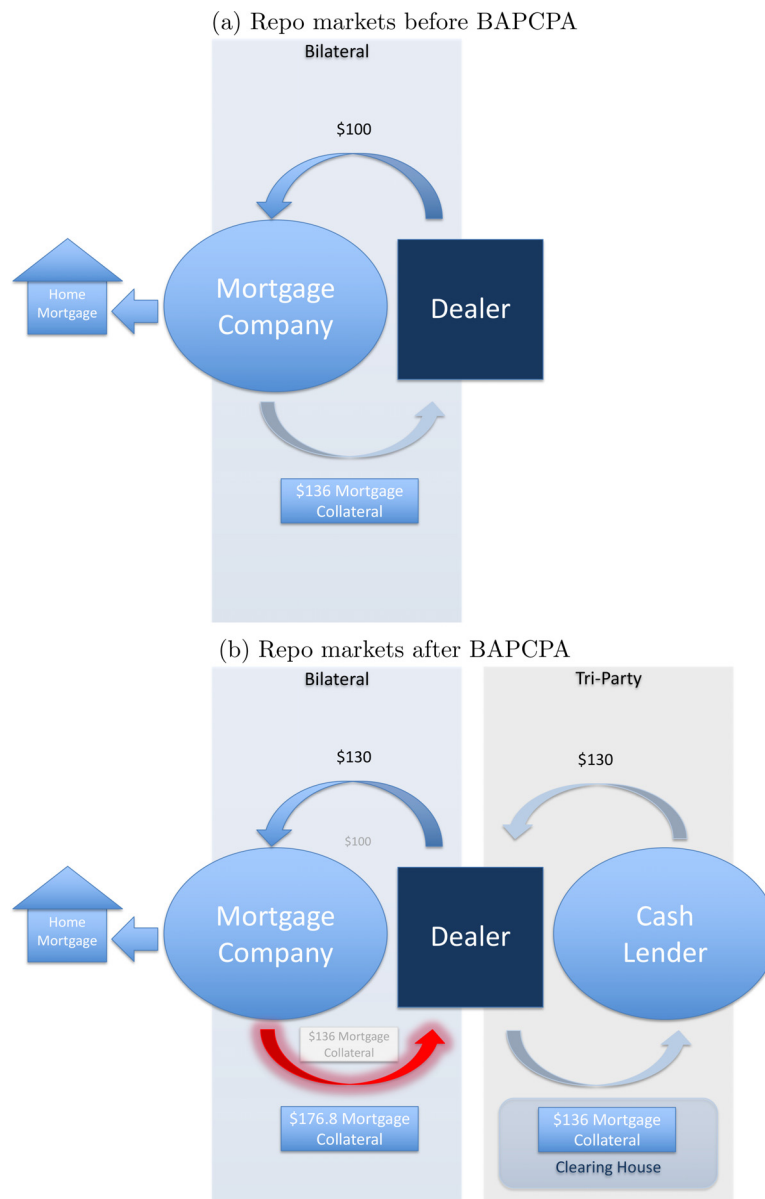


Fig. 1. REPO MARKETS BEFORE AND AFTER BAPCPA 2005.

Notes: Figures depict the process by which a dealer can borrow and lend funds via the tri-party and bilateral repurchase market respectively, using the same underlying private-label mortgage collateral provided by the cash borrower for both contracts. Figure (a) depicts the “first round” of lending from the dealer to the IMC before BAPCPA and Figure (b) depicts the proposed “second round” of lending enabled by BAPCPA.

warehoused loans quickly and pledge the newly minted securities.⁹

The literature documents, both theoretically (Gottardi et al., 2019) and empirically (Copeland et al., 2014; Infante, 2019), that differences in haircuts between the bilateral and tri-party market are larger for riskier collateral classes. The larger haircut differentials imply larger money multiplier potential. To establish the size of the haircut that dealers charged the IMCs in my data, I

utilize data from a subset of the IMCs that report both their utilization on credit lines and the value of mortgages they post. This information allows me to estimate the haircut that IMCs paid. In 2005Q4 an IMC posted a 36% haircut to insulate dealers from the IMC’s risk of default. I propose that the improved creditor rights on private-label collateral (securities or whole mortgage loans), following BAPCPA, decreased haircuts on collateral posted by the dealer in the tri-party market, from close to 100% to 5%.¹⁰

⁹ From speaking with market participants, dealers were able to securitize collateral very fast, usually several days or weeks.

¹⁰ Copeland et al. (2014) documents that the haircut charged in the tri-party market on private-label mortgage collateral was 5% in July 2008. It

Imagine that pre-BAPCPA, the dealer funded the IMC initially with \$100 of capital (the “first round” of lending), the IMC posted \$136 worth of collateral to secure the repo loan, and the dealer held the collateral. In Fig. 1, I depict this as a dealer paying \$100 to buy mortgage collateral valued at \$136, from the mortgage company in the bilateral market, with an agreement to sell it back in 60 days at \$100. The above haircuts imply that post-BAPCPA, dealers could re-sell the \$136 collateral, posted by the IMCs in the bilateral market, for \$130 in the tri-party market. The process would work as follows. If the dealer funded the IMC with the initial \$100 of capital, the dealer could borrow an additional \$130 by replying the IMC’s \$136 collateral in the tri-party repo market at a haircut of 5%. If the dealer reinvested in warehouse mortgage loans by lending the \$130 back to the IMC (the “second round” of lending), and haircuts remained the same, the IMC would post \$176.8 of new mortgages to secure the repo loan. By replying this \$176.8 of mortgage collateral, the dealer could borrow \$169 in the tri-party market. This cycle of lending to an IMC, receiving collateral from the IMC, replying the collateral to a cash lender, and reinvesting the cash in the IMC could continue for many rounds.

This haircut differential would lead to a money multiplier, similar to the multiplier in the fractional reserve system. Moreover, the dealer has incentive to reinvest each round of cash back into private-label mortgages, because the haircut differential on this collateral means that each dollar reinvested would generate more than a dollar in the dealers’ own borrowing potential, lowering the cost of capital. Additionally, although this example abstracts from an interest rate charged on the repo borrowing to simplify the exposition of overcollateralization, continuing this cycle of lending would allow the dealer to lever up the return that it received from the IMC in the form of interest rates on the warehouse repurchase facilities.¹¹

Consistent with the collateral becoming a valuable tool to lower their cost of capital, I see the dealers offering increasingly favorable overcollateralization terms to the IMCs in my data post-BAPCPA. The haircut required is 36% in 2005Q4, 26% in 2006Q1, 11% in 2006Q2, 15% in 2006Q3, and 16% in 2006Q4. The slight increase in 2006Q3 and 2006Q4 could be driven by an increase in risk of underlying mortgages. As long as the IMC generated enough money from fees or interest on the mortgages originated, it would have enough equity to fund the haircut charged by the dealer. The haircuts that I measure in 2006 are consistent with the haircut differential measured in Copeland et al. (2014), which measures the difference between median repo haircuts on private-label collateralized mortgage obligations across the bilateral and tri-party repo market in

July 2008 to be 17%. As a reference, the haircut differential on agency MBS at the time was 2%.

The haircuts I calculate for the IMCs from 2005Q4–2006Q4 allow me to calculate an upper bound on the multiplier that dealers could generate following BAPCPA. The majority of private-label mortgage securitizations sold in 30 days (Adelino et al., 2019). Therefore I use one month as the interval of the rounds of lending that generate the bilateral/tri-party haircut differential.¹² Given the quarterly haircut differentials between 2005Q4–2006Q4,¹³ 15 rounds of lending across 15 months would create a multiplier equal to:

$$1 + \sum_{i=1}^3 1.3^i + 1.3^3 \sum_{i=1}^3 1.2^i + 1.3^3 (1.2^3) \sum_{i=1}^3 1.06^i + 1.3^3 (1.2^3) (1.06^3) \sum_{i=1}^3 1.095^i + 1.3^3 (1.2^3) (1.06^3) (1.095^3) \sum_{i=1}^3 1.10^i = 66.5. \quad (1)$$

For comparison, in subsection A.8 of the Online Appendix, I illustrate that dealers could generate infinite supply of credit and leverage if haircuts remained constant.

Using the multiplier in Eq. 1, the dealer can supply \$6,650 of credit to the economy with its initial \$100 of funding. This implies that the dealer can expand its leverage secured by private-label mortgage collateral to 66.5 times its initial equity. This is 4.5 times the multiplier generated by rehypothecating Treasuries in the same way, described in subsection A.7 of the Online Appendix. This framework indicates that the credit supply increase caused by BAPCPA was larger than any credit supply increase resulting from the expansion of repo safe harbors to agency collateral in 1984 because the private-label collateral was riskier collateral and thus carried larger haircut differentials.

Decreasing the haircut that they charged the IMC would lower the money multiplier that the dealer could generate. In the limit, I would expect dealers to lower the haircuts they charged in the bilateral market until the bilateral/tri-party differential reached zero, holding constant the quality of the underlying mortgages. One potential reason that the haircut charged to IMCs did not reach the haircut charged in the tri-party market could be driven by investors realizing that the system was highly levered¹⁴ and refusing to rollover repos backed by private-label mortgage collateral. The freeze on short-term funding and interbank

is likely the haircut was 5% or lower directly following BAPCPA, since the use of private-label mortgage repo collateral was at an all time high.

¹¹ Each additional round of lending to the IMC would increase the interest earned by the dealer. The interest earned (r) on the dealer’s lending depicted in Fig. 2 (a) would be $r = \$100 \times i^{IMC}$. While in Fig. 2 (b) the dealer’s interest earned would be: $r = \$100 \times i^{IMC} + \$130 \times (i^{IMC} - i^{Dealer})$. Where i^{IMC} is the interest rate paid by the IMC on repo funding and i^{Dealer} is the interest rate paid by the dealer on repo funding.

¹² The IMC was paid when the dealer purchased the warehoused loans. If the dealer financed its purchase of the warehoused loans in the tri-party market, then once the security was sold to the final investor, the dealer would pay off the cash lender and begin another round of funding to the IMC.

¹³ The earliest that IMCs report overcollateralization is 2005Q4 and the latest is 2006Q4. Dividing one plus each of these numbers by 1.05 as in Equation 16 (in the Online Appendix), yields the haircut differential utilized.

¹⁴ In subsection A.3 of the Online Appendix, I discuss how the accounting treatment of repos pre-GFC likely allowed dealers to increase their leverage without increasing their reported leverage ratio.

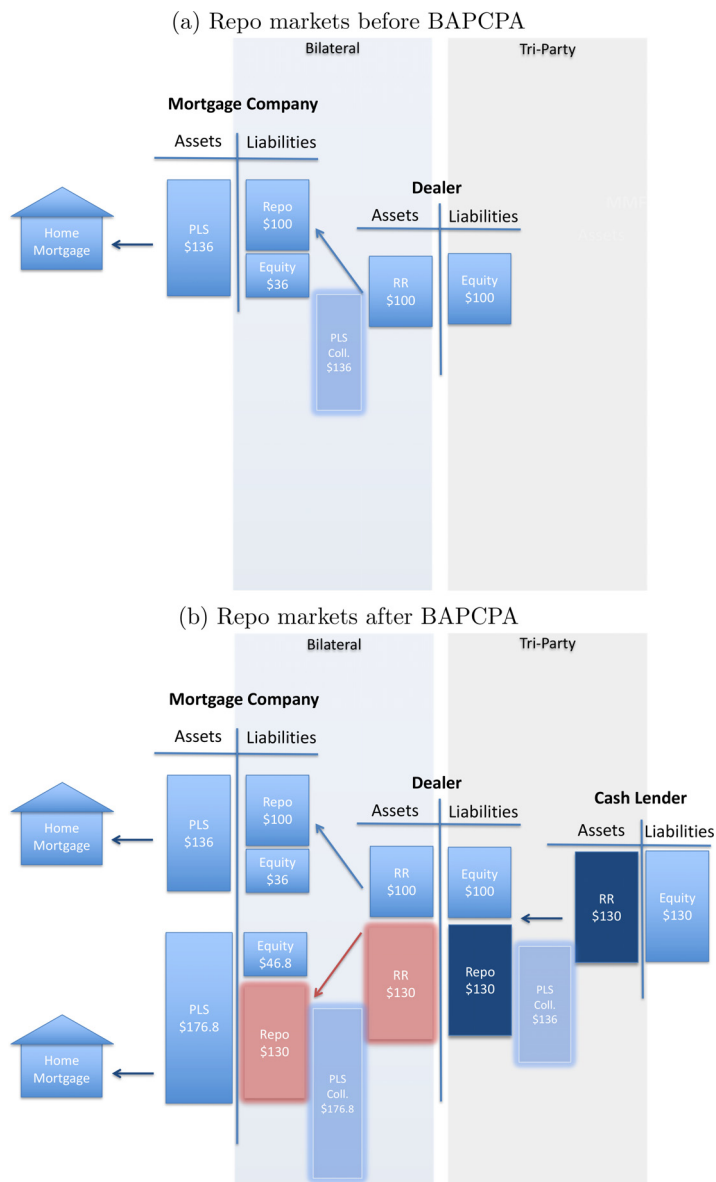


Fig. 2. REPO MONEY MULTIPLIER.

Notes: This figure depicts the increase in interlinkages, liabilities, and amount of mortgage originations resulting from BAPCPA. “RR” represents reverse repo and “PLS” indicates private-label mortgage collateral. Figure (a) depicts the “first round” of dealer lending to an IMC, possible pre-BAPCPA. Figure (b) depicts balance sheets after both a “first round” and “second round” of funding, possible post-BAPCPA. The figure abstracts from interest rates charged on repo agreements in order to focus on the money multiplier effect generated by different overcollateralization amounts charged to the IMC and to the dealer. “PLS Coll.” is private-label mortgage collateral that is posted to collateralize a repo agreement. It is not reported on the balance sheet of the dealers.

lending sparked by the funding market run on U.K. mortgage bank Northern Rock in August 2007, discussed in subsection A.9, is consistent with this narrative.

Fig. 2 depicts the effect of the repo transactions from Fig. 1 on the IMC’s, dealer’s, and cash lender’s balance sheets. The figure shows that following BAPCPA, economic credit supply and leverage increase and the number of interlinked intermediaries increases from two to four if a clearing house clears the trade for the cash lender. The first round of funding is depicted in Fig. 2 panel (a). The second round of funding, made possible by BAPCPA, is depicted in

panel (b). I depict the IMC’s second round of funding for \$130 in red because it is backed by \$176.8 of runnable collateral. If the IMC withdrew the \$176.8 of collateral from the dealer, the dealer would suffer collateral runs on its asset side as described in Infante and Vardoulakis (2021). I depict the dealer’s \$130 of repo funding from the MMF, collateralized by rehypothecated collateral, in dark blue because it is subject to repo runs on a dealer’s liability side if cash lenders refuse to roll over their funding.

If BAPCPA’s exemption from automatic stay, which allowed collateral to pass directly to the ultimate creditor in

a bankruptcy event, was not upheld, an IMC bankruptcy could cause collateral runs on a dealer's asset side. Furthermore, since each IMC borrows from multiple dealers, a failure of one IMC could cause collateral runs on multiple dealers. Indeed in its 2007 bankruptcy, American Home Mortgage, an IMC, sued its repo lenders, Credit Suisse, Bear Stearns, and Calyon, asking the court to institute the automatic stay on its mortgage collateral.¹⁵ Although BAPCPA's exemption from automatic stay was eventually upheld, the lawsuit would have frozen the collateral while the cases were decided; causing a form of collateral runs on the dealer's asset side. Additionally, one IMC failure would increase the likelihood of another since dealers respond to one IMC failure by calling margin on another (Kim et al., 2018). Consistently, the majority of the 12 IMCs that I observe declared bankruptcy in 2007, triggered by failure to meet repo margin calls.

On a dealer's liability side, BAPCPA increased the fragility of repo interlinkages by decreasing the quality of their underlying collateral and heightening the importance of cross-default clauses built into them. Cross-default clauses stated that if a repo borrower missed a margin call with one lender it was in default with all of its lenders. By granting super senior bankruptcy status to repo mortgage collateral, BAPCPA enabled creditors to sell their collateral en masse during a cross-default event, increasing the risk of fire sales. Whereas prior to BAPCPA, creditors would have needed to wait in bankruptcy court to take possession of the collateral before selling it. In this way, BAPCPA increased dealer exposure to both collateral runs on their asset side and repo runs on their liability side.

4. Expansion of repurchaseable collateral

4.1. Motivating facts

To understand whether enhancing creditor rights on the underlying mortgages increased reuse of the mortgage collateral, I first examine prices of private-label relative to agency MBS. I study the yields on both in the secondary market before and after the introduction of BAPCPA in Congress.

For MBS index i , in month t , I regress yield on the indicator variable PLS_i , which equals one for the private-label MBS index and zero for the agency MBS index, and interaction terms that interact PLS_i with monthly indicators. $\log(yield_{i,t})$ is the log of the yield on an index of MBS securities. i indicates whether the index is the LD100AS Bloomberg Barclays agency MBS index or the BNA10AS Bloomberg Barclays private-label MBS index. β_T is the coefficient of interest. It is the coefficient on the indicator variables that interact PLS_i with an indicator for each month pre and post shock. The indicator variable in January 2005 is set to zero as it was the month before BAPCPA

was introduced in Congress in February 2005.

$$\log(yield_{i,t}) = \omega Post_t + \nu PLS_i + \sum_T \beta_T PLS_i \times \mathbb{1}_{t=T} + \epsilon_{i,t} \quad (2)$$

Fig. 3 plots the coefficient β_T . The figure indicates that the yield on the PLS index decreased significantly relative to the yield on the agency MBS index following the introduction of BAPCPA in Congress, consistent with a relative increase in the price of PLS. This evidence is consistent with demand for PLS increasing after BAPCPA strengthened creditor rights on the mortgages underlying the PLS securities. Prior to the introduction of BAPCPA, PLS relative to agency yields were fairly stable. There is a slight downward trend beginning in November 2004. This may have been due to the Republicans gaining seats in the 2004 Senate elections. There had been drafts of the bill in Congress as early as 2002, however it was not thought that BAPCPA would pass until November 2004 when the Republicans gained seats in Congress.

Using dealer trading data from the FR 2004, in Fig. 4, I show evidence suggesting dealers' ability to borrow against private-label mortgage collateral tripled following BAPCPA and crashed when the repo run on MBS began in August 2007. To produce this measure, I follow Infante (2019) and calculate securities out minus securities in to proxy for the total amount of cash the dealers generated through their secured financing activities. I calculate this measure for corporate securities, the collateral class containing private-label mortgage collateral. The *securities in* reports the dollar value of lending from the dealer to participants in the bilateral market, not the total value of collateral received. When haircuts are large in the bilateral market, the true value of collateral received by dealers is not reflected in securities in. *Securities out* reports the dollar value of funding received by the dealer. Within a collateral class, if a dealer repurchased out all of the securities that it received, and used none of its own capital, securities out minus securities in would capture the value of the haircut differential that dealers were able to generate.

Subtracting securities out minus securities in gives an estimate of the amount of borrowing dealers could access by reusing private-label mortgage collateral post-BAPCPA. The increase in this measure post-BAPCPA is consistent with a large haircut differential between securities out and securities in. This large differential would allow dealers to implicitly raise private-label collateral (either whole mortgage loans or newly minted securities) in one repo market to borrow against in another. There were no significant changes that affected the other collateral classes that comprised corporate securities around the time of BAPCPA.

4.2. Empirical model

To causally test whether BAPCPA increased dealers' ability to repurchase mortgage collateral, I study the amount of repurchaseable collateral that dealers report in their annual reports from 2002 to 2006. I develop a treatment intensity difference-in-differences (DiD) research design. BAPCPA affected repo at the national level. However, my identifying assumption is that dealers who had a larger fraction

¹⁵ See Securities Industry and Financial Markets Association (SIFMA) amicus curiae brief for American Home Mortgage Holdings, Inc. v. Credit Suisse First Boston Mortgage Capital, LLC. Case No. 07-11047 (CSS) discussed further in subsection A.3 of the Online Appendix.

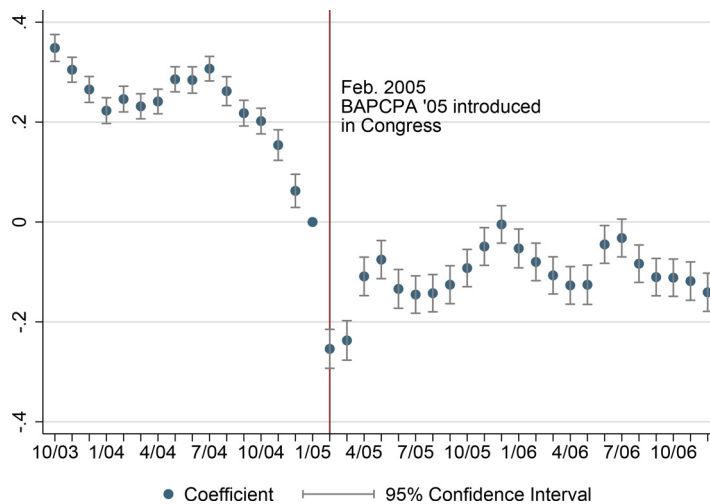


Fig. 3. PRIVATE-LABEL VS. AGENCY MBS YIELDS.

Notes: Figure plots the dynamic response of private-label MBS (PLS) relative to agency MBS yields pre vs. post the introduction of BAPCPA 2005 in Congress on February 1, 2005. I estimate Eq. 2. β_T is the coefficient of interest. It is the coefficient on the variable that interacts the indicator for PLS with an indicator for each month pre and post shock. The results indicate the yield on the PLS index dropped relative to that on the agency MBS index following the BAPCPA announcement. This is consistent with a differential increase in the price of the PLS index in February 2005 when BAPCPA was introduced in Congress.

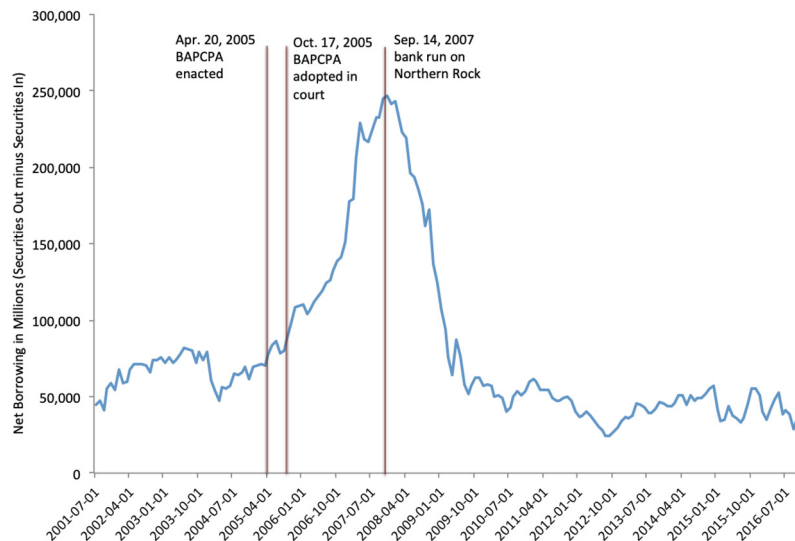


Fig. 4. DEALER NET BORROWING USING PRIVATE-LABEL MORTGAGE COLLATERAL.

Notes: Figure plots the weekly time series of dealer net borrowing backed by private-label mortgage collateral, calculated by secured borrowing (securities out) minus dealer secured lending (securities in) in the collateral class corporate securities reported in the FR 0004. In Subsection 2.2 and in the Online Appendix, I discuss the lower bound estimate of the percent that private-label mortgage collateral made up of corporate securities. Securities out includes all dealer repo and securities lending transactions. Securities in includes all reverse repo and securities borrowing transactions.

of their balance sheet exposed to private-label mortgage collateral at the time of BAPCPA would be more-affected by a strengthening of creditor rights – they were holding more illiquid collateral that suddenly became liquid. These dealers were likely to experience a greater relaxation of their leverage constraint following BAPCPA since they would have more private-label mortgage collateral available to repledge to secure their own financing.

I do not directly observe dealers' holding of private-label mortgage collateral in the mortgage warehouse at the time of the policy change so I proxy for it using dealers' underwriting of private-label MBS. I assume that deal-

ers who were securitizing more private-label MBS were also warehousing more private-label mortgages. Excerpts from the IMCs' public filings support this. For example, if a dealer was securitizing or underwriting an IMC's mortgages, it also had a "gestational line of credit" with that mortgage company.

The private-label MBS market is split into the "Alt-A" segment and the subprime segment. Subprime refers to loans given to borrowers with low credit scores (Adelino et al., 2019). The Alt-A segment is commonly referred to as "near prime," and it is typically characterized as loans to borrowers with credit scores compara-

Table 1
DEALER DESCRIPTIVE STATISTICS (2004).

	Mean (Control)	Mean (Treated)	Difference	T-stat	P-value
Number of Counties	1795	1705	90	0.14	0.890
Originations	6087	6297	–209	–0.04	0.967
N	14	6			
log(Total Assets)	20.00	20.00	0.01	0.01	0.989
log(Total Liabilities)	19.94	19.95	–0.003	–0.005	0.996
log(Total Equity)	17.04	16.92	0.12	0.23	0.817
N	16	7			

Notes: Table presents dealer descriptive statistics. Dealers in the top quartile of value of 2004 private-label MBS deals underwritten, scaled by total assets, are defined as treated dealers (scaled value of 2004 underwritten deals ≥ 0.023). Value of 2004 underwritten deals is the total value of subprime residential mortgage-backed security deals underwritten by a dealer in 2004, scaled by total assets of the dealer. Data from the CoreLogic ABS database and Inside Mortgage Finance's Mortgage Market Statistical Annual were used to compute the value of deals underwritten by a dealer.¹⁶ Total assets reports the total value of book assets in 2004Q4 for each financial institution or holding company of the financial institution. Origination and county statistics are generated using HMDA data. There are 29 (27) dealers in the reported repurchaseable collateral (within IMC, across dealer) analysis. 7 (5) of these dealers' balance sheet data I am not able to collect and I do not observe their mortgage originations in the HMDA data. These dealers all underwrote \$0 of subprime residential mortgage-backed security deals in 2004. ¹⁷ This measure was inspired by [Nadauld and Sherlund \(2013\)](#) p. 457. I am very grateful to Shane Sherlund for his help calculating this measure.

ble to those in the “prime” market, but with either income and/or assets less than fully documented or non-owner occupancy status. In 2004, subprime mortgages comprised the lion's share of the private-label MBS market ([Justiniano et al., 2017](#)). As I show in [Section 6](#) the “Alt-A” mortgage originations really began in response to BAPCPA.¹⁶

Thus I calculate PLS underwriting in the pre-period as the total value of subprime residential MBS deals underwritten and securitized by dealer in 2004. I scale the total value of deals underwritten by total book value of assets in 2004Q4 for each dealer.¹⁷

$$PLSUnderwriting_{j,2004} = \frac{\text{Total Value of Deals Underwritten}_{j,2004}}{\text{Total Assets}_{j,2004}}. \quad (3)$$

The drafts of BAPCPA as early as 2002 alleviate concerns that more- versus less- treated dealers had differential information prior to November 2004. Additionally, the securitization process generally takes several weeks to complete, indicating that the vast majority of the deals that determine the treatment variable must have been completed before November 2004.

In [Table 1](#), I present descriptive statistics showing that the more- versus less- treated dealers had similar total assets, equities, liabilities, number of mortgage originations, and originated mortgages in a similar number of counties in 2004. The heterogeneity in dealer exposure to PLS was likely driven by dealers moving into PLS in 2003 and 2004

when Fannie Mae and Freddie Mac, the two largest creators of agency MBS, were accused of accounting fraud. Regulators subsequently imposed limits on the two companies' mortgage debt holdings.¹⁸ The limits restricted Fannie Mae and Freddie Mac's creation of agency MBS, lowering barriers to entry for IMCs and private-label securitizations.

The money multiplier described in the conceptual framework implies that, by repurchasing their mortgage collateral holdings immediately following BAPCPA, more-treated dealers could support multiple rounds of new lending to an IMC. Each round would increase dealers' holdings of warehoused mortgage collateral, which they would report as repurchaseable collateral. I estimate the following DiD regression to causally identify the effect of BAPCPA on dealers' repurchaseable collateral:

$$\begin{aligned} \log(\text{RepledgeableCol}_{j,t}) &= \eta_j + \omega \text{Post}_t + \beta \text{Post}_t \\ &\quad \times PLSUnderwriting_{j,2004} + \epsilon_{j,t}. \end{aligned} \quad (4)$$

Where $\log(\text{RepledgeableCol}_{j,t})$ is the log of the repurchaseable collateral reported by dealer j at year t . Post_t is an indicator variable that equals one for 2005 and later and zero otherwise. β is the coefficient on the interaction term, $\text{Post}_t \times PLSUnderwriting_{j,2004}$, that measures how different values of PLS underwriting in 2004 affect dealers' repurchaseable collateral in the post-period relative to the pre-period. η_j contains fixed effects for each *Dealer* _{j} . The shock occurs at the dealer level so changes in repurchaseable collateral of the same dealer may be correlated. Thus I cluster at the dealer level and calculate the standard errors using the bias-adjusted cluster version of heteroskedasticity consistent standard errors. I follow the advice of [Imbens and Kolesar \(2016\)](#) and apply the “LZ2” correction to the standard errors and compute confidence intervals using a t-distribution with degrees of freedom suggested by [McCaffrey and Bell \(2002\)](#).¹⁹

¹⁶ In addition to the mortgage products discussed in [Section 6](#), I show that two-step and hybrid mortgages expanded greatly following BAPCPA, in figures available upon request.

¹⁷ This measure was taken from [Nadauld and Sherlund \(2013\)](#) p. 457 and updated with information from the CoreLogic ABS database and Inside Mortgage Finance's Mortgage Market Statistical Annual to compute the value of subprime deals underwritten by a dealer. I am very grateful to Shane Sherlund for his help calculating this measure. I scaled the value of subprime deals underwritten by each dealer by total assets of either the holding company of the dealer or the total assets of the dealer itself when a dealer was not part of a larger holding company.

¹⁸ <https://www.abcnews.go.com/Business/story?id=3664473&page=1>

¹⁹ Imbens and Kolesar present Monte Carlo evidence that the resulting confidence intervals have good coverage even with as few

Table 2
DEALER REPLEDGEABLE COLLATERAL.

	(1) log(RepledgeableCol)
Post	0.080 (0.235)
Post × PLSUnderwriting _{j,2004}	0.017* (0.008)
DealerFE	Yes
r ²	0.9440
N	90

Notes: Table reports the response of dealers' reported repledgeable collateral following BAPCPA as a function of their value of PLS deals underwritten in 2004. The independent variable, PLS deals underwritten in 2004, is scaled by dealers' total assets. I estimate the regression in Eq. 4. 19 dealers report repledgeable collateral in their financial reports. The 19 dealers are comprised of the 16 primary dealers in 2004/2005 and 3 additional dealers. The results indicate that dealers who underwrote more PLS deals in 2004 significantly increased their reported holding of repledgeable collateral following BAPCPA, consistent with a money multiplier effect.

I report the results of Eq. 4 in Table 2. A one unit increase in a dealer's underwriting of private-label MBS in 2004 significantly increased their reported repledgeable collateral by 1.7% following BAPCPA. This result is consistent with dealers who were warehousing more private-label mortgages at the time of BAPCPA being in a better position to take advantage of the large money multiplier received by pledging, at low overcollateralization rates, the warehoused or newly securitized private-label mortgage collateral. Under this interpretation, the reported increase in repledgeable collateral also functions as an indirect measure of dealers' increased lending to IMCs. The result is consistent with more-treated dealers reinvesting their increased credit supply in more private-label mortgage collateral. As they increased investment in private-label mortgage collateral by lending to IMCs – at high overcollateralization rates – dealers would receive more repledgeable collateral to report. More-treated dealers would also experience a first mover advantage relative to less-treated dealers over and above the initial burst of liquidity they experienced at BAPCPA's passage. This is because they could lend to IMCs before the haircuts charged to the IMCs began decreasing, allowing them to capitalize on the largest haircut differentials between repo markets.

5. Expansion of dealer funding to mortgage companies

There are four ways dealers could increase funding to IMCs in response to BAPCPA: (1) increasing the value of credit lines to IMCs; (2) decreasing haircuts that they required IMCs to post; (3) walking down the quality curve on the types of mortgage collateral that they fund; and (4) lowering the interest rate on their credit lines to IMCs. In this section, I focus on the maximum value of credit lines that dealers send to mortgage companies to focus on dealer credit supply and discuss the additional channels at the end of this section.

as five clusters or unbalanced cluster size. I follow code provided by Gabriel Chodorow-Reich: <https://scholar.harvard.edu/chodorow-reich/data-programs> (Chodorow-Reich et al. (2018)).

5.1. Motivating facts

I plot the average total value of warehouse credit lines extended to the IMCs in my sample pre and post shock in Fig. 5. The figure shows that prior to BAPCPA, the average value of total IMC warehouse credit lines was relatively stable around \$3 billion dollars. Post shock the average increased sharply to close to \$5 billion dollars. In Fig. 6, I present the maximum value of credit lines offered to a representative IMC. The IMC receives credit lines from four dealers, Credit Suisse, Countrywide, UBS and IXIS. According to my research design, Credit Suisse and Countrywide receive a larger credit supply increase from BAPCPA than do UBS and IXIS. Post BAPCPA, the more affected dealers, Credit Suisse and Countrywide, increased their credit lines by more than UBS and IXIS. All dealers increased or maintained their credit lines to the IMC, consistent with an overall credit supply increase.

5.2. Empirical model

In order to causally link increased ability to repledge private-label mortgage collateral to increased supply of credit, I utilize a within mortgage company, across dealer empirical strategy similar to Khwaja and Mian (2008). I exploit the fact that each of the mortgage companies receives warehouse repurchase facilities from three or more dealers simultaneously. I utilize the same research design as Section 4.2. I define treated dealers to be those whose scaled value of PLS deals underwritten in 2004, from Eq. 3, was in the top quartile:

$$\text{Treated Dealer}_j = \mathbb{1}[\text{Top Quartile PLSUnderwriting}_{j,2004}]. \quad (5)$$

The control dealers are the dealers in the bottom three quartiles.

I estimate the extent to which treated dealers increase lending to an IMC post shock, relative to control dealers lending to the same IMC, within a tight window around BAPCPA. Studying an increase in lending within a tight window allows me to isolate the effect of BAPCPA by alleviating the concern of confounding shocks occurring over the same period and by increasing the likelihood that the pre-period is a valid counterfactual for the post-period. I estimate the change in lending by treated dealers relative to untreated dealers from 2004Q3 to 2006Q3. I make the following identifying assumptions: (1) dealers who have a larger fraction of their balance sheet exposed to private-label mortgage collateral at the time of BAPCPA experience immediate ability to reuse the collateral post-BAPCPA, and (2) treated dealers have established credit lines with IMCs, which allow them to immediately pass on credit supply shocks.

The dealer lending channel (supply channel) is difficult to estimate both because BAPCPA strengthened creditor rights on collateral, which would directly increase the willingness of dealers to lend against the collateral, and because supply shocks are often correlated with demand shocks. Both supply and demand shocks would affect the dealer lending volume that I want to measure. If the dealers who receive a positive credit supply shock

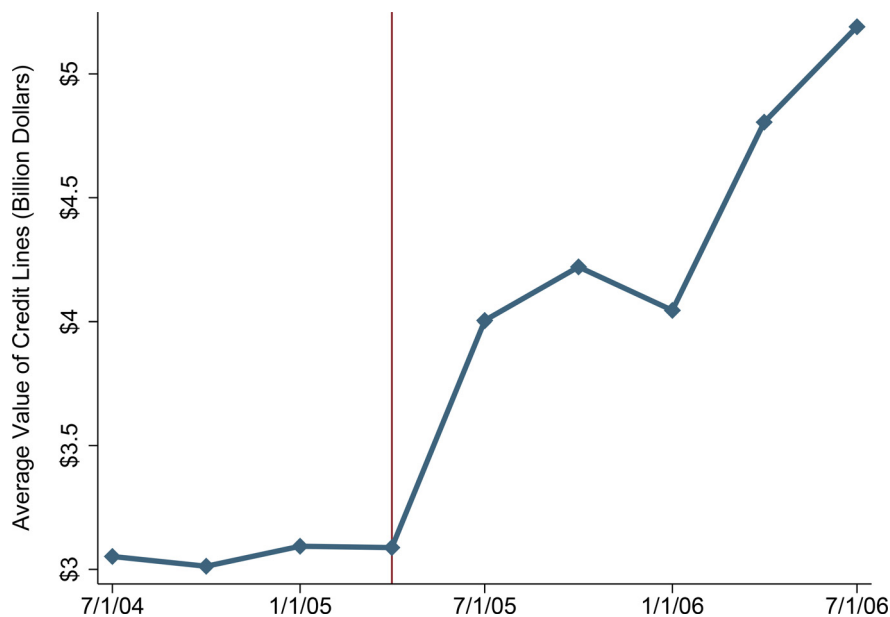


Fig. 5. AVERAGE CREDIT LINES TO MORTGAGE COMPANIES.

Notes: Figure plots the average total value of credit lines available to an IMC. Post BAPCPA, the average total credit extended to an IMC began to increase. This data is compiled from IMC quarterly filings. Figure includes all twelve IMCs in my regression analysis as well as GMAC which only reports aggregate data on the warehouse credit lines that it receives.

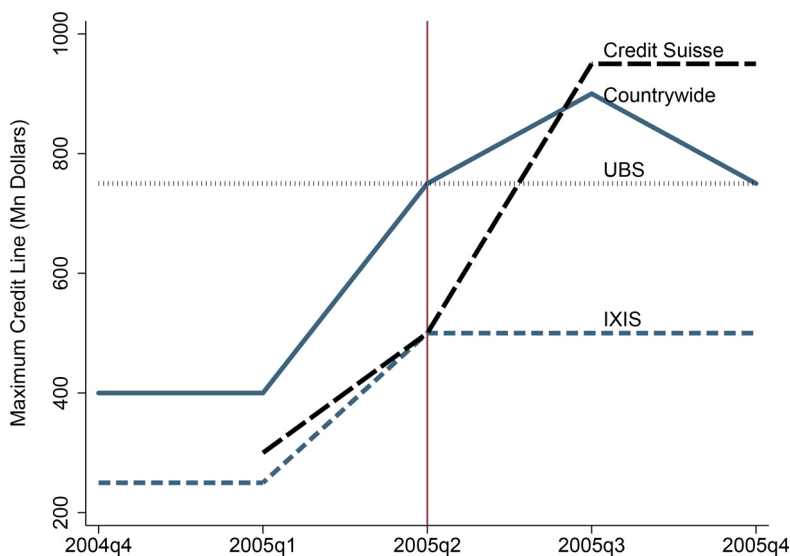


Fig. 6. MAXIMUM CREDIT LINES TO AN EXAMPLE MORTGAGE COMPANY.

Notes: The figure plots the maximum credit line values extended to an example mortgage company by dealers pre and post BAPCPA. Credit Suisse and Countrywide are treated dealers. Although all credit lines increased following BAPCPA, Credit Suisse and Countrywide increased their credit lines to the mortgage company by more than did the control dealers, UBS and IXIS.

due to BAPCPA lend more to IMCs, a concern for identification is that the IMCs to whom they lend are more productive and thus demand more credit. By studying the change in value of credit lines offered by treated dealers relative to control dealers within an IMC, I isolate the change in credit supplied to an IMC that is caused by BAPCPA's shock to the dealer's ability to reuse private-label mortgage collateral, not the IMC's demand for credit.

I estimate the difference-in-differences regression:

$$\log(\text{CreditLine}_{i,j,t}) = \gamma_{i,t} + \eta_j + \sum_T \beta_T \text{Treated Dealer}_j \times \mathbb{1}_{t=T} + \epsilon_{i,j,t}. \quad (6)$$

Where $\log(\text{CreditLine}_{i,j,t})$ is the log of the credit line extended to IMC i by dealer j in quarter t . $\text{Treated Dealer}_j \times \mathbb{1}_{t=T}$ is the interaction term between the indicator variable for treated dealers and for each quarter pre- and post-

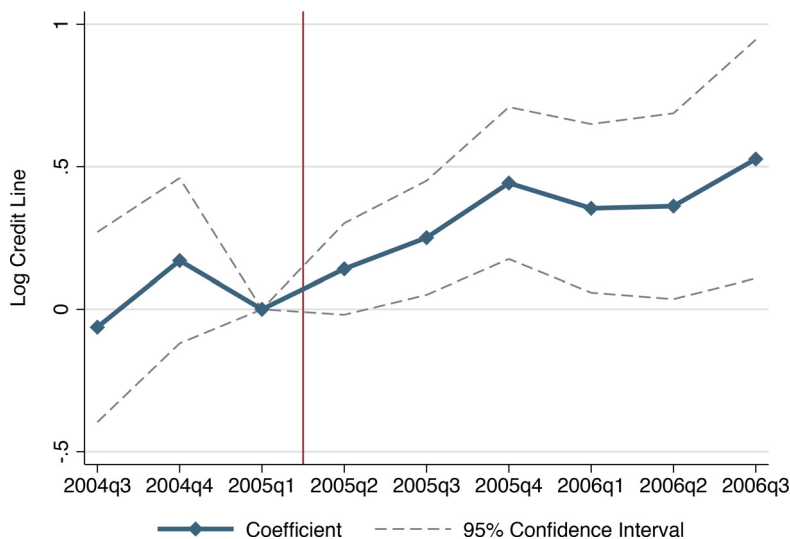


Fig. 7. EFFECT OF DEALER TREATMENT ON CREDIT LINES TO INDEPENDENT MORTGAGE COMPANIES (IMC).

Notes: Figure plots the dynamic response of treated relative to untreated dealer funding within a given IMC pre and post BAPCPA. I estimate Eq. 6. I plot the coefficient of interest, β_T . It is the coefficient on the indicator variable that interacts dealer treatment with an indicator for each quarter pre and post shock. This figure shows that prior to BAPCPA, treated and untreated dealers' lending volume to IMCs is similar. Post BAPCPA, however, the treated dealers begin to lend differentially more to IMCs.

BAPCPA. I set the reference quarter to 2005Q1, the quarter before BAPCPA was passed. β_T is the coefficient of interest, it measures the difference in lending between treated and control dealers each quarter relative to the reference quarter. I include $IMC_i \times Quarter_t$ fixed effects (FE) in $\gamma_{i,t}$ so that I can compare the lending volumes of a treated dealer to that of a control dealer both lending to the same IMC in the same quarter pre- and post-BAPCPA. The FE approach tests whether the same IMC borrowing from two different dealers experiences a larger increase in lending from the dealer who is more exposed to BAPCPA. These fixed effects absorb time-varying IMC specific factors, including IMC specific credit demand shocks. I include $Dealer_j$ FE in η_j to control for unobserved dealer heterogeneity that may be constant over time. Since the liquidity shock occurs at the dealer level, changes in credit lines from the same dealer may be correlated. I cluster the standard errors at the dealer level.

Fig. 5.2 shows the response of dealer lending volume to IMCs following BAPCPA. It presents the FE specification with a total 548 credit lines extended to the twelve IMCs from 27 dealers between 2004Q3 and 2006Q3. The results indicate a large dealer lending channel effect. Prior to BAPCPA, treated and control dealers' lending volumes within an IMC were similar. Post BAPCPA, however, the treated dealers began lending differentially more than control dealers within an IMC. Being a treated dealer is associated with a significant 29.6% increase in lending on average across the post period relative to control dealers lending to the same IMC.²⁰ The results suggest that immedi-

Table 3

WITHIN MORTGAGE COMPANY ACROSS DEALER ANALYSIS.

	(1) log(Credit Line)
Post $\times$ Treated Dealer	0.296** (0.120)
IMCxQuarterFE	Yes
DealerFE	Yes
r2	0.7005
N	548

Notes: Table reports the response of treated dealer funding relative to control dealer funding within a given IMC pre versus post BAPCPA. I estimate

$$\log(\text{CreditLine}_{i,j,t}) = \gamma_{i,t} + \eta_j + \beta \text{Post}_t \times \text{Treated Dealer}_j + \epsilon_{i,j,t}.$$

I observe 27 dealers lending to the 12 IMCs in my sample. The results are consistent with treated dealers significantly increasing their lending to a given mortgage company post BAPCPA, relative to control dealers lending to the same mortgage company in the same quarter.

ately after BAPCPA passed, dealers who were more exposed to private-label mortgage collateral prior to the shock differentially increased their lending to IMCs.

The fixed effects strategy I use does not require that dealer supply shocks and IMC demand shocks be uncorrelated. The mortgage company fixed effects will absorb any mortgage company demand shocks. One potential concern, however, is that the BAPCPA shock to dealer liquidity was anticipated so treated dealers could adjust their lending to IMCs prior to it. A benefit of my research design is that since there were early drafts of BAPCPA, treated and control dealers were equally likely to anticipate the policy change within a narrow window around BAPCPA. If the shock was anticipated, I would expect to see the treated dealers increase their lending to IMCs prior to 2005Q2.

²⁰ I run the equivalent regression to Eq. 6, however with a single pre-period and a single post-period in order to estimate the cumulative effect of BAPCPA in the post period. Post_t is an indicator variable that equals one for 2005Q2 and later and zero otherwise. Table 3 presents the results.

Table 4

ROBUSTNESS: WITHIN MORTGAGE COMPANY ACROSS DEALER ANALYSIS (PRIMARY DEALERS ONLY).

	log(Credit Line)
Post × Treated Dealer	0.373** (0.142)
IMCxQuarterFE	Yes
DealerFE	Yes
r ²	0.6881
N	401

Notes: Table reports the response of treated dealer funding relative to control dealer funding within a given IMC pre versus post BAPCPA. I estimate

$$\log(\text{CreditLine}_{i,j,t}) = \gamma_{i,t} + \eta_j + \beta \text{Post}_t \times \text{Treated Dealer}_j + \epsilon_{i,j,t}.$$

I estimate the regression for only the 15 primary dealers lending the IMCs in my sample during 2004Q3–2006Q3. There are 7 treated and 8 control primary dealers. The results are consistent with those using the entire sample of dealers lending to the mortgage companies. Treated dealers significantly increase their lending to a given mortgage company post BAPCPA, relative to control dealers lending to the same mortgage company in the same quarter.

However, the dynamic response plot of treated versus control dealer lending in Fig. 5.2 does not seem to be trending up in the pre-period. Furthermore, if there was an adjustment due to anticipation, this would bias my result downward since treated dealers would increase their lending relative to control dealers in both the pre-period and the post-period.

Another potential concern is that treated and control dealers are systematically different in ways that are not eliminated by looking at pretreatment dealer balance in Table 1. In order to alleviate this concern, I limit the analysis to just the primary dealers lending to mortgage companies. These are the 15 largest dealers who make the market for the U.S. Treasuries. These dealers likely have similar reputations and access to secured funding. Table 4 reports the results of this regression. The coefficient of interest is similar in magnitude and significance – being a treated dealer is associated with a significant 37.3% increase in lending on average across the post period relative to control dealers' lending to the same IMC. Another potential concern is that the treated dealers are dealers who underwrite more PLS deals and may therefore have incentive to originate more mortgages to capture underwriting fees post-BAPCPA. The underwriting fee incentive would not be sufficient to generate the results found in Fig. 5.2. I provide

further discussion of underwriting fees in the subsection A.4 of the Online Appendix.

5.2.1. Testing credit supply expansion hypothesis

In this section, I test whether the increased supply of credit by treated dealers is more consistent with a substitution effect or with an overall credit supply increase following BAPCPA. To establish this, I break the twelve IMCs into two groups. “Treated IMCs” are defined as the six IMCs that receive an above median fraction of their warehouse credit lines from treated dealers prior to 2005Q1. “Control IMCs” are defined as the six IMCs that receive a below median fraction of their warehouse credit lines from treated dealers prior to 2005Q1. In Table 5, I present descriptive statistics showing that the treated and control IMCs had similar total assets,²¹ number of mortgage originations, and originated mortgages in a similar number of counties in 2004.

I define $\text{CreditLine}_{i,t}$ as the sum of credit lines that a mortgage company receives from all dealers, j , that it is linked to in a given quarter:

$$\text{CreditLine}_{i,t} = \sum_j \text{CreditLine}_{i,j,t}. \quad (7)$$

In Eq. 8, I regress $\log(\text{CreditLine}_{i,t})$ for a given mortgage company, i , in quarter, t , on an interaction term between an indicator variable equal to one in the post period and an indicator equal to one for *Treated IMC* _{i} . I include mortgage company fixed effects, γ_i , and quarter fixed effects, α_t . There are twelve IMCs and I cluster the standard errors at the IMC level. As in Eq. 4, I calculate the standard errors using the bias-adjusted cluster version of heteroskedasticity consistent standard errors and apply the “LZ2” correction to the standard errors, following Imbens and Kolesar (2016):

$$\log(\text{CreditLine}_{i,t}) = \beta \text{Post} \times \text{Treated IMC}_i + \gamma_i + \alpha_t + \epsilon_{i,t}. \quad (8)$$

The results in Table 6 suggest treated mortgage companies receive a significant 19.9% increase in total maximum credit available in the post period relative to control mortgage companies. This evidence suggests that BAPCPA led to an increase in overall lending to mortgage companies with an above median fraction of their credit lines from treated dealers, rather than a substitution of lending

²¹ In addition to total assets, treated and control IMCs did not have statistically significant differences in $\log(\text{TotalEquity})$ and $\log(\text{TotalLiabilities})$, tables are available upon request.

Table 5

INDEPENDENT MORTGAGE COMPANY (IMC) DESCRIPTIVE STATISTICS (2004).

	Mean (Control)	Mean (Treated)	Difference	T-stat	P-value
Number of Counties	1683	2000	−317	−0.54	0.602
Originations	4010	3527	483	0.27	0.796
log(Total Assets)	14.6	14.6	0	0.04	0.97
N	6	6			

Notes: Table presents descriptive statistics for independent mortgage companies (IMCs) collected from quarterly filings and HMDA data. IMCs with an above median fraction of their total credit lines from treated dealers in 2005Q1, the quarter before BAPCPA was passed, are defined to be treated (≥ 0.55). Total assets reports the total value of book assets in 2004Q4 for each IMC. Origination and county statistics are generated using HMDA data. There are 12 IMCs in my dataset.

Table 6
TREATED IMC CREDIT LINES.

	(1) log(Credit Line)
Post × Treated IMC	0.199** (0.072)
IMCFE	Yes
r ²	0.9484
N	102

Notes: Table reports the total value of credit lines received pre versus post BAPCPA as a function of whether or not the mortgage company was treated. I estimate Eq. 8. Treated dealers are those who receive an above median fraction of their credit lines from treated dealers in 2005Q1, the quarter before BAPCPA was passed. The result suggests that treated IMCs receive a significant increase in total credit supplied relative to control IMCs post BAPCPA.

away from control toward treated dealers within a mortgage company. If BAPCPA caused IMCs to substitute away from control dealers toward treated dealers without increasing their total credit supply, there would be no statistically significant increase in total credit lines for treated IMCs. This is a lower bound of the overall credit supply increase created by BAPCPA. It only captures the increase in credit to treated IMCs over and above that of control IMCs. However, all dealers were eventually affected, which would increase the credit supply to both the treated and control IMCs.

My results suggest that in response to BAPCPA, dealers expanded credit for risky mortgage products – consistent with the strengthened creditor rights allowing collateral quality to decrease while leaving its expected recovery value the same. Dealers loosened the covenants that they imposed in the form of funding sublimits on their credit lines. These sublimits specified the maximum amount of the credit line that could be allocated to fund certain types of mortgage loans. A subset of the twelve IMCs report the funding sublimits specified by their dealers. As plotted in Fig. 8, the sublimits for interest-only, second-lien, jumbo, non-owner occupied, and 120–180 day past due loans doubled post BAPCPA. I do not observe dealers increasing credit lines for conforming mortgage products following BAPCPA. In subsection A.4 in the Online Appendix, I report sublimits on additional mortgage products and wet, or implicitly unsecured funding, by dealer.

6. Mortgage company lending

In this section, I study whether IMCs pass the credit supply increase along to households. Due to data limitations discussed in the data section, I conduct a county level treatment intensity analysis. I create the variable $IMCMarketShare_{c,2004}$ to capture the exposure of a county to the IMCs in 2004. I calculate this variable using the number of mortgage originations in the HMDA data:²²

²² I construct the 2004 IMC county market share using value of mortgage originations and find that the distribution of market share is very similar to the measure using number of originations.

$$IMCMarketShare_{c,2004}$$

$$= \frac{\text{Number of originations by IMCs}_{c,2004}}{\text{Total number of all originations}_{c,2004}}. \quad (9)$$

My preferred specification includes all originations by IMCs in the numerator. As discussed in the data section, the 12 IMCs originate the majority of all mortgages originated by IMCs in 2006 and engage in correspondent lending to other IMCs. The 12 IMCs each received the majority of their funding from overlapping subsets of the 29 largest dealers. Since these dealers were the largest, most interconnected, they were likely lending to the IMCs for which I am not able to collect data. Indeed, Stanton et al. (2014) find that dealer funded warehouse repurchase facilities dominate the IMC funding model. While the research design in Table 4.2 and Table 5.2 exploits heterogeneity in dealers' initial treatment intensity, all dealers were eventually exposed to BAPCPA; furthermore the research design utilizes both a narrow window around BAPCPA. Thus it estimates a lower bound of the credit supply increase since all dealers experienced an increased ability to reuse collateral that persisted longer than the window studied. The increased collateral reuse would increase all dealer credit supply and incentivize them to increase warehouse lending to IMCs, thereby affecting the vast majority of IMCs. In subsection B.3 of the Online Appendix, I present a robustness test that calculates the $IMCMarketShare_{c,2004}$ variable using just originations by the six IMCs with an above median fraction of total funding from the most-treated dealers and my results persist.

Fig. 9 depicts all IMC market share per county in 2004. The variation in IMC market share is likely due to the Fannie Mae and Freddie Mac scandals in 2003 and 2004 which decreased barriers to entry for IMCs to enter the mortgage market. Justiniano et al. (2017) and Drechsler et al. (2021) show that events in 2003 led to a sudden surge in the private-label mortgage market.²³ The market share of IMCs however was relatively stable throughout 2004, the year that I define my treatment. To alleviate concerns that my results are driven by a pre-existing trend in IMC expansion, I examine pre-period home prices and income and include *state × month* and *county* fixed effects. In my home price analysis, I study a longer pre-period – almost two years, June 2003 to March 2005 – and establish pre-treatment parallel trends in home prices for treated and control counties. If an increased presence of IMCs in the mortgage market drove up home prices, it should be reflected in a pre-treatment increase in home prices in the treated counties in 2003 when IMCs' market share was surging. I also find no statistically significant difference in the 1999 census per capita income in counties with high versus low IMC market share after controlling for state fixed effects. The pre-period balance in home prices and income helps to mitigate concerns that the treatment and control counties were significantly dif-

²³ In subsection B.5, I find that although the market share of IMCs was rising during 2000 to 2003, the steep decline in agency mortgage originations in 2003 coincides with the increased relative share of IMC (and therefore private-label) originations. This is consistent with the Fannie/Freddie fraud cases restricting Fannie/Freddie market share.

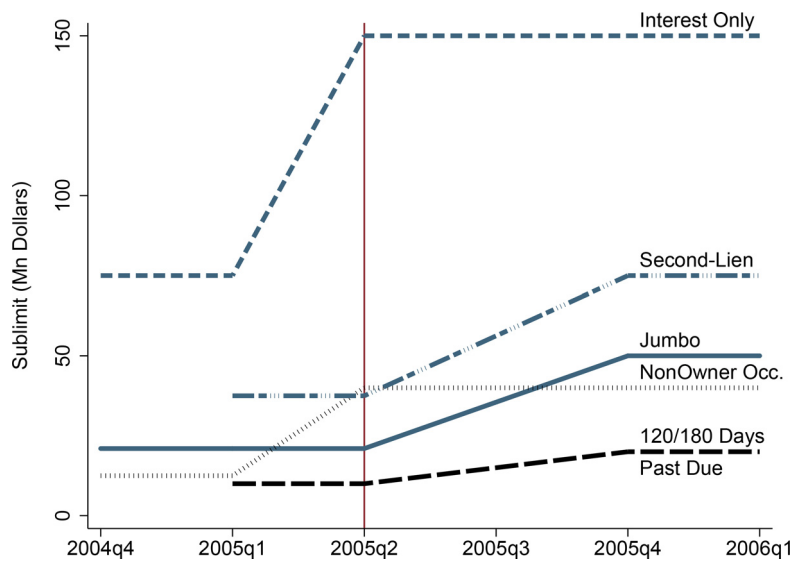


Fig. 8. MAXIMUM CREDIT LINES TO AN EXAMPLE MORTGAGE COMPANY.

Notes: The figure plots the sublimit of funding available to fund certain mortgage products. The dealers would specify the maximum amount of funding per credit line available to fund certain mortgage products. The figure suggests that following BAPCPA dealers increased funding for riskier mortgage products.

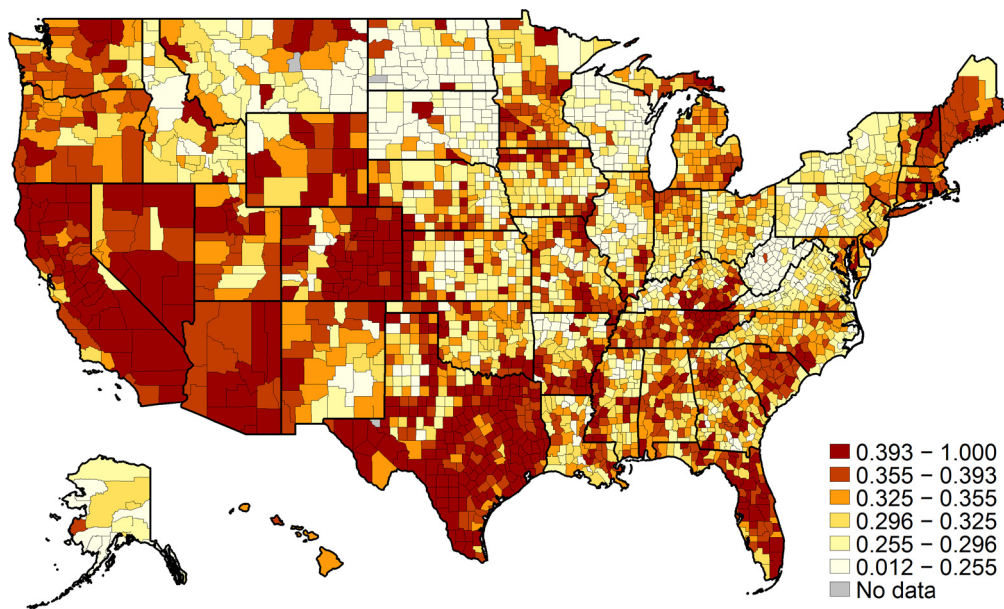


Fig. 9. INDEPENDENT MORTGAGE COMPANY (IMC) MARKET SHARE.

Notes: The figure depicts the county level market share of all IMCs reported in 2004. The market shares are calculated using the 2004 HMDA data.

ferent in ways that varied with BAPCPA after controlling for fixed effects.

I investigate how $IMCMarketShare_{c,2004}$ affects county level mortgage characteristics and home prices. I run the following dynamic regression:

$$Y_{c,t} = \gamma_c + \eta_{s,t} + \sum_T \beta_T IMCMarketShare_{c,2004} \times \mathbb{1}_{t=T} + \epsilon_{c,t}. \quad (10)$$

Where $Y_{c,t}$ is the variable of interest in county c , at month t , γ_c denotes county level fixed effects, and $\eta_{s,t}$ denotes state $\times$ month fixed effects. $IMCMarketShare_{c,2004} \times \mathbb{1}_{t=T}$ is

the interaction term between the market share variable, and an indicator variable for month of origination.²⁴ The reference month is March 2005, the month prior to the passage of BAPCPA. Standard errors are clustered at the county level. I report the regression results with county fixed effects alone and with both county and state $\times$ month fixed effects. The regression with both county and state $\times$

²⁴ I estimate the analogous regression, with a single pre and post period, for all independent variables studied in Eq. 10 and report the results also in Table 7.

Table 7
IMC COUNTY MARKET SHARE EFFECT ON MORTGAGE CHARACTERISTICS.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	log(Orig)		BalloonFrac		log(IntlntRt)	All IMCs Affected	HzdRt		log(hpBoom)		log(hpBust)	
$Post \times IMCMarketShare_{c,2004}$	0.375*** (0.013)	0.268*** (0.080)	0.005*** (0.001)	0.030*** (0.004)	0.175*** (0.006)	-0.239*** (0.033)	0.331*** (0.058)	0.141*** (0.044)	0.443*** (0.094)	0.209** (0.082)	-0.333*** (0.106)	-0.329*** (0.114)
CountyFE	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
StateMonthFE	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
r ²	0.9642	0.9947	0.1507	0.5239	0.8543	0.9482	0.0402	0.0447	0.9768	0.9957	0.9702	0.9919
N	8728	8572	9000	8874	9000	8874	355,154	355,134	19,232	18,929	15,831	15,628

Notes: Table reports the response of housing market characteristics in a given county as a function of the 2004 market share of independent mortgage companies (IMCs) in that county. I run the regression

$$Y_{c,t(t)} = \gamma_c + \eta_{s,t} + \beta Post_t \times IMCMarketShare_{c,2004} + \epsilon_{c,t(t)}$$

In county, c at month t . All dependent variables except the default hazard rate are measured at the county, month level. The default hazard rate (Y_t) regression is estimated at the loan level. Y_t is calculated as an indicator variable equal to one if the loan ever defaults and zero otherwise. At the county level, the specification measures the fraction of loans originated 5 months prior to April 2005, that ever defaulted, and compares it to the fraction originated just post April 2005, that ever defaulted, as a function of all IMC market share. γ_c represents county level fixed effects, $\eta_{s,t}$ represents state $\times$ month fixed effects, $IMCMarketShare_{c,2004}$ is the IMC county level market share in a given county in 2004, the year before the shock occurs. β is the coefficient of interest. It is the coefficient on the interaction between $IMCMarketShare_{c,2004}$ and the post period. This coefficient measures the change in the dependent variable if $IMCMarketShare_{c,2004}$ increased from 0% to 100%. I use the Public HMDA data to compute the 2004 county level IMC market share and the county month HMDA data to study originations.^a I use CoreLogic LLMA data to study mortgage characteristics and Zillow's ZHVI to study home prices.

^aNeil Bhutta publishes the HMDA data reported at the county month level on his personal website: <https://www.sites.google.com/site/neilbhutta/data>.

month fixed effects is my preferred specification as this regression compares mortgage characteristics in counties with high versus low IMC market shares within the same state and month, absorbing state-month housing market effects. For all of the mortgage characteristic regressions, I study a narrow window around BAPCPA. The narrow window, fixed effects, and stable pre-period IMC market share help to ensure that the pre period is a valid counterfactual for the post period. Callaway et al. (2021) note that continuous treatment in a DiD setting can introduce bias if the estimator's weighting of treatment doses differs from the population-weighted treatment doses. In subsection B.2 in the Online Appendix, I show that the population-weighted doses are close to normally distributed and therefore the estimator closely approximates the population weights, thus the potential bias would be small.

The first dependent variable that I study is $\log(Originations_{c,t})$. Originations include both refinance mortgages, and purchase mortgages.²⁵ Fig. 10 (a) plots the evolution of the coefficient of interest, β_T – the effect of the interaction term between origination month and $IMCMarketShare_{c,2004}$ – from September 2004 to February 2006. The figure shows that originations in counties that had a higher IMC market share in 2004 were not statistically different from those in less exposed counties prior to the policy change. A 10% increase in total IMC market share in 2004 leads to a significant 2.7% increase in total mortgage originations on average in the post period. The slight initial drop in originations following BAPCPA is due to IMCs' transition from subprime mortgage products to alternative mortgage products.

Alternative products offered more potential to expand originations given their ability to lower initial monthly payments by using low introductory interest rates and negative amortizing features. These alternative mortgage products were classified as prime by CoreLogic. Prior to BAPCPA, the fraction of prime and subprime originations moved in parallel for high versus low IMC market share counties. Fig. 10 (b) shows that following BAPCPA, counties with a higher market share of IMCs first began originating a significantly higher fraction of subprime and lower fraction of prime mortgages, before rigorously expanding into the prime market. This transition led to the slight relative decrease initially, followed by the robust expansion in originations in high IMC counties. It likely took a few months following BAPCPA for IMCs to pivot into alternative products as they had never been originated in such volume prior to 2005. This evidence is consistent with an already saturated subprime market limiting opportunities for credit supply expansion. In the subsection B.1, in the Online Appendix, the Federal Reserve's Loan Officer Survey suggests that mortgage demand prior to BAPCPA was declining.

I study the effect of BAPCPA on these non-traditional, alternative (Alt-A), mortgage products that were near prime in borrower credit score but had riskier payment

²⁵ As a robustness test, I estimate purchase and refinance originations using the HMDA data. I also estimate all analyses using CoreLogic mortgage originations aggregated to the county month level. Results remain consistent and are reported in the Online Appendix or on request.

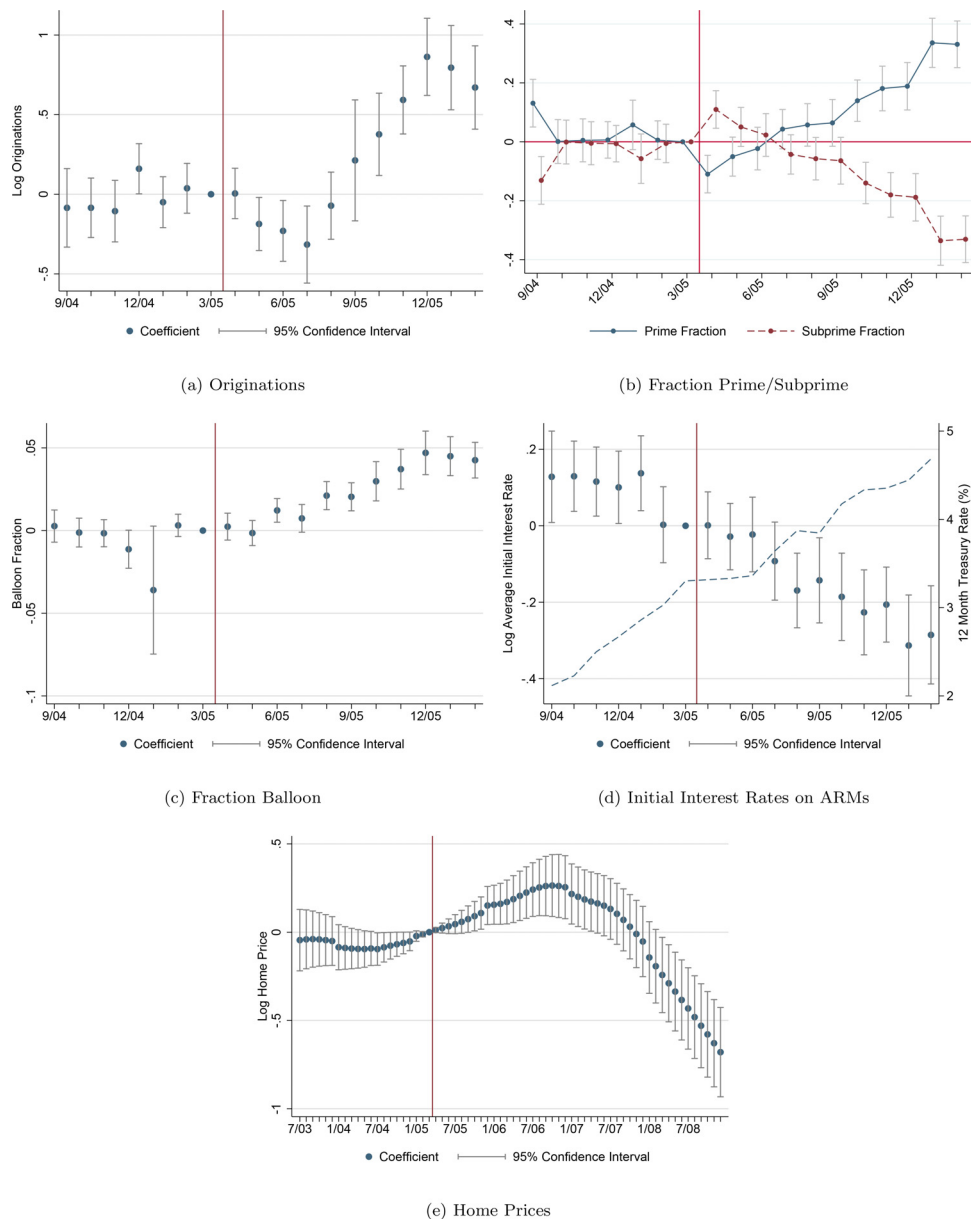


Fig. 10. IMC COUNTY MKT SHARE EFFECT ON MTG CHARACTERISTICS.

Notes: Figures plot the dynamic response of mortgage characteristics in a given county to the 2004 IMC market share in that county. I estimate Eq. 10. β_T is the coefficient of interest. It is the coefficient on the variable that interacts $IMCMarketShare_{c,2004}$ with month. I use the public HMDA data to compute the 2004 county level IMC market share, CoreLogic and the county month HMDA data to study originations, and Zillow's ZHVI to study home prices.

schedules. Balloon mortgages, including ARM balloons, do not fully amortize over the term of the loan. They leave a large balance or balloon payment due at maturity. Borrowers of these mortgages are more likely to experience negative equity when home prices stop rising. Negative equity decreases ability to refinance and the borrower may not have the resources to pay off the balance at the end of the loan even if the home is sold. Gerardi et al. (2017) find that negative equity is a quantitatively important factor in default for strategic motives.

I estimate the dynamic regression in Eq. 10, where the dependent variable is fraction of balloon mortgages. I calculate the fraction of balloon mortgages relative to the to-

tal number of mortgages originated in that county in a given month. Fig. 10 (c) shows that prior to BAPCPA, the fraction of balloon mortgages originated in counties with higher IMC market share was not statistically different from that in less exposed counties. A 10% increase in total IMC market share in 2004 leads to a significant 0.3 percentage point increase in the fraction of balloon mortgages originated in that county on average in the post period. Similarly, I find that the fraction of negative amortizing mortgages increases and the fraction of owner-occupied mortgage originations decreases in counties more exposed to IMCs following BAPCPA (see Table 10 in the Online Appendix).

I also estimate Eq. 10 to study the effect of $IMCMarketShare_{c,2004}$ on introductory mortgage interest rates. I limit the sample to only ARM originations and study the average initial interest rates charged in a county pre- and post-BAPCPA. In Fig. 10 (d), I plot the response of the $\log(Initial\ Interest\ Rate_{c,t})$, along with the twelve-month Treasury rate, to which IMC financial reports state ARMs were pegged during this period. Prior to BAPCPA, there was no statistical difference in the average initial interest rate charged on mortgages between counties with high and low market share of IMCs. Post-BAPCPA, a 10% increase in total IMC market share leads to a significant 2.39% decrease in the average initial interest rate. The twelve-month Treasury rate over this period increased monotonically. This evidence is consistent with mortgage companies originating mortgages with low initial “teaser” rates. These rates did not reflect the interest payment required to fully amortize the loan but rather an artificially low rate advertised to attract potential borrowers. The rates reset to the actual interest rate after a specified point in time, increasing the risk of “payment shock” to the borrower.

The results suggest that BAPCPA’s super senior creditor rights on loans in the mortgage warehouse led dealers to expand funding for alternative mortgage products as shown in Fig. 8 and the IMCs to increase originations of them. The results suggest these products lowered near-term mortgage payments at the peak of the housing boom to increase demand when conforming mortgages were expensive. Due to their payment structure however, these products were vulnerable to home price declines and interest rate increases.

If the loans originated post-BAPCPA by IMCs are riskier, the default hazard rate should increase in counties with higher IMC market share as of 2004. To test this, I limit the dataset to loans that were originated from November 2004 to September 2005, a five-month window around the passage of BAPCPA. I set the indicator variable $Defaulted\ Loan_i$ equal to one if the loan ever enters 90 day delinquency, foreclosure, or becomes an REO property in its lifetime and zero if the loan remains active. I run the following regression:

$$Defaulted\ Loan_i = \gamma_c + \eta_{s,t} + \beta\ Post_t \times IMCMarketShare_{c,2004} + \epsilon_i. \quad (11)$$

A 10% increase in IMC market share significantly raises the hazard rate on mortgages originated post shock by 1.4 percentage points. I report the regression results for Eq. 11 in Table 7 (columns labeled HzdRt). This table also reports the regressions estimated in Eq. 10 with a single pre- and post-period, to understand the overall effect of BAPCPA on the housing market. I find a 9.1% increase in IMC originations in response to BAPCPA.²⁶ I utilize my estimate of the default hazard rate to estimate a 70% implied

marginal hazard rate on mortgages originated in response to BAPCPA. Applying the marginal hazard rates to the increase in originations, BAPCPA accounts for 38% of defaults among all loans originated during 2005 and 2006. I describe these calculations in subsection B.6 of the Online Appendix.

This increase in defaults has implications for our understanding of the GFC because one key feature of the crisis was an unexpected level of mortgage defaults. The results support the increase in alternative products found in Justiniano et al. (2017); the finding in Albanesi et al. (2017) that mortgage defaults during the crisis were concentrated in the middle and top of the credit score distribution; the finding in Demyanyk and Van Hemert (2011) that mortgages originated in 2006 and 2007 performed worse than mortgages originated between 2000–2004 after controlling for borrower characteristics; as well as the result in Ospina and Uhlig (2018) that prime MBS performed worse than subprime MBS. These events are not likely to be unique to mortgages. If commercial real estate (CRE) loans or collateralized loan obligations (CLOs) were granted preferred bankruptcy status in repo markets, there would likely be a similar chain of events. Private originators, facing a fixed supply of creditworthy borrowers, would originate products with riskier amortization structures or lower credit quality to meet increased capital-market demand for the collateral.

To understand whether increased mortgage originations drove up home prices by increasing the demand for homes, I estimate Eq. 10 where the dependent variable is $\log(HomePrice_{c,t})$ over the period from June 2003 to December 2008. Fig. 10 (e) shows the dynamic response of home prices in a county to the 2004 market share of IMCs in that county. Prior to BAPCPA, IMC market share in 2004 was not associated with a differential change in home prices. Post-BAPCPA, a 10% increase in market share leads to a significant 2.1% increase in home prices between April 2005 and November 2006 followed by a steep and significant decline in home prices from January to December 2008. A 10% increase in 2004 market share leads to a significant 3.3% decrease in home prices during 2008, relative to their pre-period level. This evidence is consistent with the rise in defaults among loans originated post-BAPCPA as well as with the wave of IMC bankruptcies caused by margin calls on their repo credit lines. The results suggest that the credit supply increase drove up demand for homes in the counties most exposed to alternative products and initially masked their fragility.

In Appendix C in the Online Appendix, I present a natural extension of the Gertler and Kiyotaki (2015) model that features anticipated bank runs. In the model, I increase banks’ operational advantage consistent with dealers’ increased ability to reuse collateral following BAPCPA. In response, the model predicts that dealers increase their leverage and the amount of credit supplied to the economy. Relative to the baseline model, this amplifies the price increase of capital in a boom and its decline in a bust, consistent with the empirical results of this paper. Increasing banks’ operational advantage also increases the probability of a bank run in every state, which is consistent with the GFC of 2008.

²⁶ I multiply 26.8%, the estimated increase in mortgage originations caused by a 100% increase in IMC market share (Table 7), by the total market share of IMCs in the pre-period, which was 34%. This is an underestimate of IMC market share since it is calculated using the HMDA data, which does not account for mortgage purchases from correspondent lenders.

7. Conclusion

This paper provides three main contributions. First, it establishes that improved creditor rights increase the reuse of collateral in the repo markets. It then establishes empirically that when overcollateralization requirements differ at each leg of the chain of reuse, increased collateral reuse functions like a money multiplier, creating a positive credit supply shock. Finally, this paper documents that the pass-through of the credit supply increase from the repo market to the housing market, created by BAPCPA, was economically large, contributing to both the housing boom and its bust in the GFC.

Declaration of Competing Interest

I, the author, declare that there is no conflict of interest.

Data availability

The code for this article is available at <https://doi.org/10.17632/vygrym46ph.2>.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.jfineco.2023.06.001](https://doi.org/10.1016/j.jfineco.2023.06.001)

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